

AWS Services Keyword Cheat Sheet

How to use: Find the phrase that matches what the exam question says, match it to the keywords, and identify the answer. 533 entries across 40 categories.



Cost & Billing

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Analyze spending over time	cost trends, spend history, charts, graph, forecast, monthly bill, usage report, visualization	AWS Cost Explorer	Built for viewing and analyzing cost and usage data with graphs and forecasts.
See where money is going	cost breakdown, service spend, linked account, region, tag, filter costs, allocate costs	AWS Cost Explorer	Breaks down and filters spend data by service, region, tag, or account.
Predict future spend	forecast, projected cost, future cost, estimate usage, predict bill	AWS Cost Explorer	Forecasting future costs is a classic Cost Explorer signal.
Get alerts before overspending	budget, threshold, notification, alert, overage, spending limit, email alert, SNS alert	AWS Budgets	Budgets is built for setting spending alerts and limits with notifications.

Set spending guardrails	max spend, warning, threshold, limit, cap spending, zero spend, cost limit	AWS Budgets	Budgeting and notification are the defining clues.
Estimate monthly cost before launch	pricing calculator, forecast before launch, estimate cost, what will it cost, monthly estimate	AWS Pricing Calculator	Estimation <i>before</i> deployment points to Pricing Calculator, not Cost Explorer.
Find savings recommendations	best practice, recommendation, optimize, reduce waste, rightsizing, underutilized, cost saving	AWS Trusted Advisor	Trusted Advisor gives improvement and optimization recommendations.
Detect unused resources	idle, underutilized, unused, rightsize, wasted, over-provisioned, zombie resource	AWS Trusted Advisor	Rightsizing and waste reduction are Trusted Advisor clues.
Check service limits	service limits, quota, account limits, limit check, approaching limit, request limit increase	AWS Trusted Advisor	Trusted Advisor checks limits and warns about risks.
Compare support plans	basic, developer, business, enterprise, phone support, response time, TAM, technical account manager	AWS Support Plans	Support level and response time questions point to Support Plans.

Manage many accounts centrally	organizations, consolidated billing, governance, master account, payer account, SCP, member account	AWS Organizations	Multiple accounts and central billing point to Organizations.
Apply policies across accounts	SCP, organizational unit, OU, restrict services, prevent actions, organization-wide policy	AWS Organizations (SCPs)	SCPs let you restrict what services member accounts can use.
Tag resources for cost allocation	cost allocation tag, tag policy, resource tag, billing tag, tag-based report	AWS Cost Allocation Tags	Tags must be activated for Cost Explorer to group costs by tag.



Monitoring & Logging

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Monitor metrics and alarms	metrics, alarms, logs, dashboard, CloudWatch, threshold alarm, log group, log stream, latency	Amazon CloudWatch	Monitoring keywords point to CloudWatch. Also covers log aggregation and dashboards.
Record API activity / audit	API calls, audit, history, tracking, who made changes, account activity, CloudTrail, governance log	AWS CloudTrail	CloudTrail logs account-level API activity for auditing.

Log resource configuration changes	configuration history, drift, compliance, config rule, resource inventory, change timeline	AWS Config	Configuration tracking and compliance point to AWS Config, not CloudTrail.
Detect threats automatically	threat detection, unusual API calls, malware, anomaly, suspicious activity, compromised instance	Amazon GuardDuty	GuardDuty uses ML to detect threats across CloudTrail / VPC Flow / DNS logs.
Scan for vulnerabilities	vulnerability scan, CVE, patch finding, container image scan, EC2 scan, software vulnerability	Amazon Inspector	Inspector scans EC2 and container images for known CVEs.
Automate responses to events	event rule, cron, schedule, trigger Lambda, event bus, EventBridge, on-schedule, event-driven automation	Amazon EventBridge	EventBridge routes events and triggers automations on schedule or event.
Collect logs from many sources	log aggregation, centralized logging, log ingestion, log agent, CloudWatch agent, unified logging	Amazon CloudWatch Logs	All AWS service logs can be funneled into CloudWatch Log Groups.
Trace requests across services	X-Ray, distributed tracing, service map, latency trace, end-to-end trace, performance bottleneck	AWS X-Ray	X-Ray traces requests through microservices / Lambda for debugging latency.
Receive personal health alerts	service health, planned maintenance, degraded	AWS Health Dashboard	Personal Health Dashboard shows events

service, Health Dashboard,
AWS Health API

that affect your specific
account.



Identity & Security

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Manage permissions	user, role, policy, access, permission, IAM, grant access, least privilege, principal, resource policy	AWS IAM	IAM handles identity, authentication, and access control for AWS resources.
Temporary access / assume role	assume role, temporary credentials, cross-account, STS, federated, short-lived token, AssumeRole	IAM Role (STS)	Temporary access always means a Role, not a User. STS issues the credentials.
Secure root account	root user, MFA, avoid daily use, root credentials, delete root keys, root best practice, enable MFA	IAM Root Best Practice	Root account protection is a recurring IAM exam theme.
Service-to-service access	service access, EC2 to S3, Lambda to DynamoDB, cross-service, execution role, instance profile	IAM Role	Cross-service access always uses a Role, not an access key.

Store secrets and credentials	password, secret, API key, rotation, database credential, secret store, Parameter Store	Secrets Manager / SSM Parameter Store	Secrets + automatic rotation = Secrets Manager. Static config values = Parameter Store.
Encrypt data / manage keys	encryption, key management, KMS, CMK, customer master key, envelope encryption, encrypt at rest	AWS KMS	Key management and encryption clues point to KMS.
Patching responsibility question	who patches OS, who secures host, customer vs AWS, shared responsibility, managed service patching	Shared Responsibility Model	Any "who is responsible for X?" question is a Shared Responsibility Model clue.
Protect web apps from attacks	WAF, web application firewall, SQL injection, XSS, cross-site scripting, rate limit, HTTP filter, bot	AWS WAF	WAF filters malicious HTTP/S traffic before it reaches CloudFront or ALB.
Protect against DDoS attacks	DDoS, volumetric attack, Shield, layer 3, layer 4, SYN flood, amplification attack	AWS Shield	Shield Standard is free; Shield Advanced adds 24/7 DDoS response team.
Single sign-on / federated login	SSO, single sign-on, SAML, Active Directory, federated identity, corporate credentials, IAM Identity Center	AWS IAM Identity Center	Lets employees use corporate credentials to access AWS — replaces AWS SSO.

User sign-up and login for apps	user pool, app authentication, OAuth, OIDC, social login, JWT, Cognito, app user directory	Amazon Cognito	Cognito handles auth for your application's end users (not AWS console users).
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Centralized firewall management	firewall policy, network firewall, centralized rules, inspection VPC, east-west traffic	AWS Network Firewall	Network Firewall provides stateful deep-packet inspection across VPC traffic.
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Networking

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Protect internet traffic (instance level)	firewall, allow/deny traffic, inbound, outbound, port, security group, stateful, instance firewall	Security Group	Security Groups are stateful instance-level firewalls. Port rules are the giveaway.
Control subnet traffic	subnet-level rule, stateless, NACL, network ACL, deny subnet, allow CIDR, subnet firewall	Network ACL (NACL)	NACLs are stateless and operate at the subnet level — the subnet clue is key.
Isolate networks / VPC	VPC, private subnet, public subnet, VPC peering, CIDR, internet gateway, NAT gateway, network isolation	Amazon VPC	VPC is the network isolation layer. Subnets, IGW, and NAT live inside VPCs.

Hybrid connection (private/dedicated)	on-premises, private connection, direct link, dedicated bandwidth, Direct Connect, colocation	AWS Direct Connect	Direct Connect = private dedicated line between on-prem and AWS. Not over the internet.
Encrypted VPN over internet	encrypted tunnel, site-to-site, VPN, IPsec, customer gateway, virtual private gateway, client VPN	AWS VPN	VPN = encrypted tunnel over the public internet. Direct Connect is the private alternative.
Load balance traffic	distribute traffic, multiple targets, high availability, ALB, NLB, listener, target group, health check	Elastic Load Balancing (ELB)	Balancing and distributing traffic across targets points to ELB.
Scale automatically	traffic changes, add/remove instances, scaling group, auto scale, scale out, scale in, launch template	Auto Scaling	Auto scale and scaling group are the giveaway terms.
DNS and domain routing	domain name, hostname, route traffic, DNS, Route 53, latency-based, failover, weighted, geolocation	Amazon Route 53	DNS + routing clues point to Route 53. Global traffic routing with health checks too.
Faster content delivery / CDN	cache, edge, low latency, global users, CDN, CloudFront, edge location, geo-restrict, signed URL	Amazon CloudFront	Edge caching and CDN are the giveaway. CloudFront = global content delivery network.

Connect VPCs without peering complexity	Transit Gateway, hub and spoke, centralized routing, many VPCs, cross-region VPC, on-prem to many VPCs	AWS Transit Gateway	Transit Gateway acts as a central hub routing traffic between VPCs and on-prem.
Private endpoint to AWS services	VPC endpoint, PrivateLink, private S3 access, no internet, interface endpoint, gateway endpoint	VPC Endpoint / PrivateLink	VPC Endpoints let resources access AWS services privately without going through internet.
Accelerate global app performance	global accelerator, Anycast, fixed IP, global entry point, latency routing, TCP/UDP acceleration	AWS Global Accelerator	Routes traffic through AWS backbone for lower latency globally.



Compute

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Elastic virtual server	instance, VM, virtual machine, compute, EC2, launch instance, AMI, key pair, instance type, t2 t3 m5	Amazon EC2	EC2 is AWS's virtual server (IaaS). Any instance/VM language points here.
Serverless function	no server, event-driven, run code, Lambda, function, trigger,	AWS Lambda	Lambda is the serverless compute

	invocation, serverless, FaaS		service. No server management required.
Container orchestration	containers, cluster, tasks, pods, Kubernetes, ECS, EKS, container management, Docker, microservices	ECS / EKS	ECS = AWS-native containers. EKS = managed Kubernetes. Both run containers.
Serverless containers	no server, container runtime, pay per use, Fargate, serverless container, no EC2, managed container infra	AWS Fargate	Fargate = serverless containers. No EC2 management; pay per task.
Deploy app with minimal ops	platform, deploy code, managed deployment, Elastic Beanstalk, PaaS, zip upload, auto deploy, worker tier	AWS Elastic Beanstalk	Beanstalk handles infra. You just upload code. Simpler than EC2 + manual setup.
Define infrastructure as code	template, repeatable setup, stack, automated deployment, CloudFormation, IaC, YAML template, rollback	AWS CloudFormation	Templates and stacks always point to CloudFormation.
Deploy and manage code releases	deployment group, rolling update, CodeDeploy, blue/green, canary, in-place deployment, deploy pipeline	AWS CodeDeploy	CodeDeploy automates deployments to EC2, Lambda, or on-prem servers.

Build CI/CD pipeline	pipeline, source, build, test, deploy, CodePipeline, CodeBuild, continuous integration, continuous delivery	AWS CodePipeline / CodeBuild	CodePipeline orchestrates CI/CD stages; CodeBuild compiles and tests code.
Run batch workloads	batch job, compute job, managed batch, job queue, job definition, HPC, parallel workload	AWS Batch	AWS Batch runs batch computing jobs at scale without managing infra.
Run code at the edge	edge function, Lambda@Edge, CloudFront function, viewer request, origin request, edge logic	Lambda@Edge / CloudFront Functions	Runs code closer to users at CloudFront edge locations for ultra-low latency.



Storage

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Store files / objects	bucket, object, upload, download, lifecycle, S3, put object, key, prefix, static, versioning, replication	Amazon S3	Buckets and objects are S3. Also used for static website hosting.
Archive data cheaply	archive, long-term retention, infrequent access, Glacier, cold	S3 Glacier / Deep Archive	Archive + long-term = Glacier. Deep Archive is

	storage, tape replacement, vault		cheapest with slowest retrieval.
Block storage for EC2	volume, snapshot, disk, attached storage, EBS, gp2, gp3, io1, persistent volume, boot volume	Amazon EBS	EBS = block storage volumes attached to EC2. Snapshots are stored in S3.
Shared file storage	file system, shared access, NFS, EFS, POSIX, mount, shared drive, multi-AZ file, Linux file share	Amazon EFS	EFS = shared NFS file system. Multiple EC2 instances can mount it simultaneously.
Physical data migration	migrate large data, ship device, offline transfer, Snowball, petabyte, physical appliance, snowmobile	AWS Snowball / Snowmobile	Physical device transfer points to Snowball. Snowmobile for exabyte-scale.
Windows file share in cloud	SMB, Windows file system, Active Directory integration, FSx, Windows Server share	Amazon FSx for Windows	FSx for Windows = fully managed Windows-native file shares with AD integration.
High-performance file system	HPC, Lustre, high throughput, scratch file system, FSx for Lustre, ML training data	Amazon FSx for Lustre	FSx for Lustre = high-performance file system for ML and HPC workloads.
Continuously sync on-prem to cloud	DataSync, online transfer, sync on-premises, NFS SMB to S3, scheduled sync, incremental transfer	AWS DataSync	DataSync automates and accelerates moving data between on-prem and AWS storage.

Back up
resources
across services

backup policy, centralized
backup, backup vault,
recovery point, AWS
Backup, cross-region
backup

AWS Backup

AWS Backup provides
centralized backup across
EC2, RDS, EFS,
DynamoDB, and more.



Database

**If the
question
says...**

Look for these keywords

Answer is...

Why this is the clue

SQL /
relational
database

relational, SQL, schema,
tables, joins, RDS, MySQL,
PostgreSQL, Oracle,
Aurora, ACID

Amazon RDS

Managed relational
database. Aurora is
RDS-compatible but
significantly faster.

NoSQL /
key-value
database

key-value, low latency,
flexible schema,
DynamoDB, NoSQL,
document, item, partition
key, sort key, GSI

Amazon DynamoDB

DynamoDB = serverless
NoSQL. Key-value and
low-latency are the
giveaways.

Cache
database
data

in-memory, faster reads,
low latency, ElastiCache,
Redis, Memcached,
session store,
sub-millisecond

Amazon ElastiCache

In-memory caching + speed
= ElastiCache. Redis
supports persistence;
Memcached is simpler.

Managed graph database	graph, relationships, nodes, edges, Neptune, social network, fraud detection, knowledge graph	Amazon Neptune	Neptune is a fully managed graph database. Nodes and relationships are the clues.
Time-series database	time series, IoT data, timestamps, metrics over time, Timestream, sensor data, time-stamped readings	Amazon Timestream	Timestream is purpose-built for time-series data like IoT telemetry.
Fully managed ledger database	immutable, cryptographic, ledger, QLDB, audit trail, tamper-proof, blockchain-like	Amazon QLDB	QLDB provides an immutable cryptographically verifiable transaction log.
Migrate databases to AWS	DMS, database migration, schema conversion, SCT, heterogeneous migration, homogeneous migration	AWS DMS / SCT	DMS migrates databases; SCT converts schema for heterogeneous migrations.
Global multi-region database	global tables, DynamoDB global, Aurora global, multi-region active-active, global replication	DynamoDB Global Tables / Aurora Global	Multi-region active-active write capability — DynamoDB for NoSQL, Aurora for SQL.



Analytics

If the question says...	Look for these keywords	Answer is...	Why this is the clue
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Data warehouse / analytics	analytics, OLAP, business intelligence, Redshift, data warehouse, BI tool, columnar, Tableau, petabyte query	Amazon Redshift	Warehousing and OLAP analytics keywords point to Redshift.
Stream data in real time	streaming, real-time, kinesis, data stream, firehose, ingestion, event stream, clickstream, IoT data	Amazon Kinesis	Real-time data streaming and ingestion point to Kinesis.
Search and analyze logs	search logs, full-text search, Elasticsearch, OpenSearch, log analytics, dashboards, Kibana, index	Amazon OpenSearch Service	Full-text search + log analytics = OpenSearch (formerly Elasticsearch).
Connect data sources / ETL	ETL, glue, data catalog, transform, pipeline, job, crawler, extract transform load, data lake	AWS Glue	ETL and data catalog keywords point to Glue. Crawler discovers schema automatically.
Query S3 data with SQL	Athena, query S3, serverless query, analyze CSV Parquet JSON, no database, pay per query	Amazon Athena	Athena lets you run SQL directly on S3 files — no database needed.
Visualize data / BI dashboards	QuickSight, dashboard, visualization, BI, SPICE, business intelligence tool, charts, reports	Amazon QuickSight	QuickSight is AWS's managed BI and visualization service.
Build a data lake	data lake, lake formation, centralized repository, raw	AWS Lake Formation	Lake Formation simplifies building, securing, and

data store, structured
unstructured data

managing data lakes on
S3.

Real-time data
processing
with Kafka

MSK, managed Kafka,
Apache Kafka, producer
consumer, pub sub,
streaming topics

Amazon MSK

MSK is managed Apache
Kafka. Kafka / pub-sub /
topics are the clues.



Application Integration

**If the question
says...**

Look for these keywords

Answer is...

Why this is the clue

Decouple
microservices with
queue

SQS, queue, message,
decoupled, producer
consumer, buffering, async,
polling, visibility timeout

Amazon SQS

SQS queues decouple
producers from
consumers for async
message processing.

Fan-out
notifications to
many

SNS, topic, publish
subscribe, fan-out, push
notification, email SMS alert,
pub-sub

Amazon SNS

SNS pushes messages
to many subscribers at
once — fan-out is the
keyword.

Coordinate
distributed
workflows

Step Functions, state
machine, orchestration,
workflow, task coordination,
saga pattern, retry logic

**AWS Step
Functions**

Step Functions
orchestrates multi-step
workflows with retries
and error handling.

Build REST or
HTTP APIs

API Gateway, REST API,
HTTP API, endpoint, rate
limit, throttle, API key,

**Amazon API
Gateway**

API Gateway creates,
manages, and secures

	WebSocket, resource method		APIs for backend services or Lambda.
Send transactional emails	SES, email, send email, SMTP, transactional email, marketing email, bounce, complaint	Amazon SES	SES is AWS's email sending service for transactional and marketing emails.
Connect apps with event-driven architecture	EventBridge, event bus, SaaS events, partner event source, schema registry, event routing	Amazon EventBridge	EventBridge routes events from AWS services, SaaS partners, and custom apps.



Machine Learning

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Build and train ML models	SageMaker, model training, Jupyter, notebook, ML pipeline, train deploy, feature store, model registry	Amazon SageMaker	SageMaker is the end-to-end ML platform for building, training, and deploying models.
Add AI without ML expertise	Rekognition, Comprehend, Translate, Polly, Transcribe, pre-trained model, AI service, no-code AI	AWS AI Services	Pre-built AI services add vision, NLP, translation, and speech without training models.
Recognize images or video	Rekognition, face detection, object detection, image	Amazon Rekognition	Rekognition analyzes images and videos for

	analysis, label detection, celebrity recognition		faces, objects, scenes, and text.
Convert speech to text or text to speech	Transcribe, Polly, speech, voice, text-to-speech, speech recognition, audio transcription	Amazon Transcribe / Polly	Transcribe = speech-to-text. Polly = text-to-speech.
Translate languages	Translate, language detection, multilingual, real-time translation, batch translation	Amazon Translate	Translate provides fast, accurate neural machine translation.
Extract text from documents	Textract, OCR, document processing, form extraction, table extraction, handwriting, PDF parsing	Amazon Textract	Textract goes beyond OCR — it extracts structured data from forms and tables.
Build conversational chatbots	Lex, chatbot, voice bot, NLU, intent, slot, conversation flow, Alexa-like	Amazon Lex	Lex provides the NLU and ASR technology behind Alexa for building chatbots.



Cost & Billing — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
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Receive credits or discounts for committed usage	reserved instance, RI, savings plan, committed use, 1-year, 3-year, upfront, partial upfront, no upfront	Reserved Instances / Savings Plans	Committing to 1 or 3 years gives up to 72% discount vs on-demand.
Use spare capacity at a discount	spot instance, spot fleet, interruption, bid price, fault-tolerant workload, spot market	EC2 Spot Instances	Spot instances use unused EC2 capacity — cheapest compute but can be interrupted.
Dedicated physical server for compliance	dedicated host, physical server, BYOL, bring your own license, per-socket licensing, compliance isolation	EC2 Dedicated Hosts	Dedicated Hosts give you a physical server — needed for per-socket/per-core licensing.
Run workloads on dedicated hardware	dedicated instance, single tenant, hardware isolation, regulatory requirement	EC2 Dedicated Instances	Dedicated Instances run on hardware not shared with other AWS customers.
Get cost anomaly alerts	anomaly detection, unexpected spike, cost anomaly, unusual spend, cost alert, AI-driven alert	AWS Cost Anomaly Detection	Uses ML to detect unusual spending patterns and alert you automatically.
View free tier usage	free tier, 12 months free, always free, free usage limit, free tier alert	AWS Free Tier / Budgets	Free tier usage is tracked in the Billing console; Budgets can alert when approaching limit.

Centralize purchasing of reservations

RI sharing, reservation sharing, centralized RI, savings plan sharing, organizational RI

AWS Organizations (RI Sharing)

Reserved Instance and Savings Plan benefits can be shared across linked accounts.



Monitoring & Logging — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Create custom metrics	custom metric, put metric data, application metric, business metric, custom namespace	Amazon CloudWatch (Custom Metrics)	You can publish any metric to CloudWatch using the PutMetricData API.
Alert when metric crosses threshold	alarm, SNS notification, auto scaling trigger, breached threshold, OK ALARM INSUFFICIENT	Amazon CloudWatch Alarms	CloudWatch Alarms watch a single metric and trigger actions when breached.
Retain and query operational logs	log retention, query logs, log insights, CloudWatch Logs Insights, filter pattern, log query	CloudWatch Logs Insights	Logs Insights lets you query log data using a purpose-built query language.
Track changes to AWS resource configs	who changed, what changed, configuration snapshot, compliance timeline, resource history	AWS Config	Config records resource configurations and evaluates them against rules.

Check if resources comply with rules	compliance check, config rule, non-compliant, remediation, conformance pack	AWS Config Rules	Config Rules evaluate resources against desired configurations automatically.
Get notified about AWS service outages	status page, service disruption, region outage, availability zone issue, service event	AWS Service Health Dashboard	Shows real-time status of all AWS services — public facing unlike Personal Health Dashboard.
Collect metrics from on-premises servers	hybrid monitoring, on-prem metrics, CloudWatch agent, server monitoring, non-AWS resource	CloudWatch Agent	CloudWatch Agent can run on on-prem or non-AWS servers to send metrics and logs.
Correlate security findings in one place	SIEM, security findings, Security Hub, aggregated findings, CIS benchmark, FSBP	AWS Security Hub	Security Hub aggregates findings from GuardDuty, Inspector, Macie, and more.
Detect sensitive data in S3	Macie, PII, sensitive data discovery, S3 data classification, personal data, GDPR scan	Amazon Macie	Macie uses ML to discover and classify sensitive data (PII, credentials) in S3.



Identity & Security — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
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Manage access across multiple AWS accounts	cross-account access, switch role, assume role cross-account, trust policy, account B access	IAM Cross-Account Roles	Cross-account access always uses a Role with a trust policy — never share credentials.
Grant external identity access	identity provider, IdP, SAML 2.0, OpenID Connect, web identity, federated user	IAM Identity Federation	Federated identities from an external IdP assume an IAM Role to get temporary credentials.
Rotate access keys automatically	key rotation, access key age, credential rotation, programmatic access, rotate keys	IAM Access Key Rotation / Secrets Manager	Regular rotation is IAM best practice; Secrets Manager automates it for RDS/other secrets.
Enforce MFA across organization	MFA requirement, multi-factor authentication, virtual MFA, hardware MFA, TOTP	IAM MFA	MFA adds a second factor — exam often asks about enabling MFA for root and IAM users.
Manage SSL/TLS certificates	certificate, SSL, TLS, HTTPS, ACM, certificate manager, auto-renew, domain validation	AWS Certificate Manager (ACM)	ACM provisions and auto-renews public SSL/TLS certificates for use with AWS services.
Protect against account compromise	anomalous login, root activity, credential exposure, unauthorized access, GuardDuty finding	Amazon GuardDuty	GuardDuty detects account-level threats like credential compromise and privilege escalation.

Inspect and filter network traffic	deep packet inspection, intrusion detection, IDS/IPS, network firewall, stateful inspection	AWS Network Firewall	Network Firewall provides fine-grained traffic inspection inside your VPC.
Scan code for security vulnerabilities	code scan, SAST, static analysis, CodeGuru, security recommendations, code review	Amazon CodeGuru Security	CodeGuru reviews code and finds security vulnerabilities using ML.
Control access to S3 bucket from specific VPC	bucket policy, VPC condition, aws:SourceVpc, S3 bucket policy restriction	S3 Bucket Policy with VPC Condition	Bucket policies can restrict access to requests coming from a specific VPC or VPC endpoint.
Hardware security module	HSM, hardware security, CloudHSM, FIPS 140-2, dedicated HSM, cryptographic hardware	AWS CloudHSM	CloudHSM provides dedicated hardware for key storage — stricter than KMS.



Networking — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Route traffic based on URL path	path-based routing, /api, /images, content-based routing, ALB rules, host-based routing	Application Load Balancer (ALB)	ALB routes based on URL path or host header — Layer 7 intelligence.

Route traffic based on IP/TCP	TCP load balancer, UDP, extreme performance, NLB, static IP, low latency TCP	Network Load Balancer (NLB)	NLB operates at Layer 4 — ultra-high performance with static IPs.
Use a single IP for global app	static anycast IP, global fixed IP, single entry point, Global Accelerator, performance routing	AWS Global Accelerator	Provides two static IPs that route traffic over AWS backbone to optimal region.
Allow EC2 in private subnet to access internet	NAT gateway, NAT instance, private subnet internet, outbound only, egress	NAT Gateway	NAT Gateway allows private subnet resources to initiate outbound internet connections.
Block specific IP addresses	block IP, deny IP, NACL deny, WAF IP block, IP set, blocklist	Network ACL or AWS WAF	NACLs block IPs at subnet level; WAF blocks IPs at the application layer.
Enable communication between two VPCs	VPC peering, peer connection, inter-VPC, private routing between VPCs	VPC Peering	VPC Peering privately connects two VPCs. Not transitive — use Transit Gateway for many.
Access internet from VPC	internet gateway, IGW, public subnet, public IP, internet access	Internet Gateway (IGW)	IGW is attached to a VPC to enable public subnet resources to reach the internet.
Resolve private DNS in VPC	Route 53 Resolver, private hosted zone,	Route 53 Resolver	Resolver handles DNS for resources inside VPCs

	split-horizon DNS, VPC DNS, inbound endpoint		including private hosted zones.
Connect on-prem DNS to AWS DNS	Route 53 Resolver outbound endpoint, conditional forwarder, hybrid DNS, on-prem DNS resolution	Route 53 Resolver Endpoints	Inbound/outbound endpoints allow DNS query forwarding between on-prem and AWS.
Reduce latency with local AWS infrastructure	Local Zone, ultra-low latency, metropolitan area, local deployment, single-digit millisecond	AWS Local Zones	Local Zones place AWS compute/storage closer to large population centers.
Run AWS services on-premises	Outposts, on-premises AWS, local AWS hardware, rack, hybrid cloud hardware	AWS Outposts	Outposts brings AWS services and infrastructure physically into your data center.
Deliver content to users in specific countries	geo-restriction, country block, CloudFront geo, whitelist country, blacklist country	CloudFront Geo Restriction	CloudFront can allow or block content delivery based on viewer's country.



Compute — Extended

If the question says...

Look for these keywords

Answer is...

Why this is the clue

Deploy containers without managing clusters	App Runner, no cluster, source code to container, auto deploy, managed containers, web app container	AWS App Runner	App Runner deploys containerized web apps directly from source or image — no cluster setup.
Run scheduled or recurring jobs	cron job, scheduled task, recurring Lambda, EventBridge schedule, periodic job	EventBridge Scheduler / Lambda	EventBridge Scheduler triggers Lambda or other targets on a cron or rate schedule.
Deploy across multiple Availability Zones	multi-AZ, high availability, fault tolerant, AZ spread, availability zone redundancy	Multi-AZ Deployment (Auto Scaling + ELB)	Multi-AZ with Auto Scaling and ELB is the standard HA pattern for EC2 workloads.
Use existing VMware workloads on AWS	VMware, vSphere, vCenter, VMware Cloud on AWS, lift and shift VMware	VMware Cloud on AWS	Lets you run VMware workloads on dedicated AWS hardware without refactoring.
Store and share custom machine images	AMI, golden image, custom image, EC2 image builder, image pipeline, base image	EC2 Image Builder / AMI	Image Builder automates creation and patching of AMIs.
Manage configuration of EC2 fleets	Systems Manager, SSM, patch baseline, run command, session manager, fleet management	AWS Systems Manager	SSM manages EC2 instances at scale — patching, commands, inventory, and baselines.

Connect to EC2 without SSH keys	Session Manager, SSM Session Manager, browser-based shell, no bastion host, no open port 22	AWS Systems Manager Session Manager	Session Manager provides secure shell access without opening inbound ports or SSH keys.
Build serverless REST APIs	Lambda + API Gateway, serverless backend, API + function, no server API	Lambda + API Gateway	The classic serverless API pattern: API Gateway handles HTTP, Lambda runs business logic.
Use prebuilt software on EC2	Marketplace, AMI Marketplace, third-party software, commercial AMI, one-click deploy	AWS Marketplace	Marketplace offers pre-configured AMIs and SaaS products from third-party vendors.



Storage — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Move infrequently accessed data to cheaper tier	S3 Intelligent-Tiering, lifecycle policy, infrequent access tier, S3-IA, transition rule	S3 Lifecycle Policies / S3 Intelligent-Tiering	Lifecycle policies automate tiering; Intelligent-Tiering does it automatically without lifecycle rules.
Protect S3 objects from	versioning, MFA delete, object lock, WORM,	S3 Versioning / Object Lock	Versioning retains all versions; Object Lock

accidental deletion	accidentally deleted, recover deleted object		prevents deletion for a set retention period.
Replicate S3 data across regions	CRR, cross-region replication, SRR, same-region replication, S3 replication, disaster recovery S3	S3 Cross-Region Replication (CRR)	CRR automatically replicates objects to a bucket in another region for DR or compliance.
Access S3 as a local file system	S3 File Gateway, Storage Gateway, on-prem NFS to S3, hybrid storage, file gateway	AWS Storage Gateway (File Gateway)	File Gateway exposes S3 as a local NFS/SMB mount for on-premises applications.
Use tape backups but move to cloud	Tape Gateway, virtual tape library, VTL, backup software, Veeam, Commvault to cloud	AWS Storage Gateway (Tape Gateway)	Tape Gateway replaces physical tape with virtual tapes stored in S3 and Glacier.
Access block storage from on-premises	Volume Gateway, iSCSI, block storage gateway, on-prem block, cached volume, stored volume	AWS Storage Gateway (Volume Gateway)	Volume Gateway presents cloud-backed iSCSI block volumes to on-prem servers.
Store container images	ECR, container registry, Docker image, image repository, push pull image, container image store	Amazon ECR	ECR is the managed Docker container registry — integrated with ECS and EKS.
Store and version code	CodeCommit, Git, source control, repository, private Git, managed Git	AWS CodeCommit	CodeCommit is AWS's managed Git-compatible source control repository.



Database — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Automatically scale database capacity	Aurora Serverless, auto pause, variable load, unpredictable traffic, scale to zero DB	Amazon Aurora Serverless	Aurora Serverless automatically scales compute up/down and can pause when idle.
Failover database automatically	multi-AZ RDS, automatic failover, standby replica, RDS failover, synchronous replication	Amazon RDS Multi-AZ	Multi-AZ keeps a synchronous standby — automatic failover in 60–120 seconds.
Read-heavy database workload	read replica, read scaling, reporting workload, offload reads, asynchronous replica	RDS Read Replicas	Read Replicas scale read throughput. They are asynchronous — not for failover.
High-performance relational database	Aurora, MySQL-compatible, PostgreSQL-compatible, 5x faster, Aurora cluster, writer reader	Amazon Aurora	Aurora is MySQL/PostgreSQL compatible but up to 5x faster than standard RDS MySQL.
Cache results from relational DB	DAX, DynamoDB Accelerator, microsecond latency, in-memory DynamoDB, DAX cluster	Amazon DAX	DAX is an in-memory cache purpose-built for DynamoDB — microsecond response times.

Hybrid database (on-prem + cloud)	RDS on Outposts, database on Outposts, on-premises RDS, local database AWS managed	RDS on AWS Outposts	RDS can run on Outposts for managed relational databases in your data center.
Store JSON documents	DocumentDB, MongoDB-compatible, document store, JSON, MongoDB workload, NoSQL document	Amazon DocumentDB	DocumentDB is a MongoDB-compatible managed document database for JSON data.
Low-cost key-value at massive scale	DynamoDB on-demand, pay per request, no provisioned capacity, unpredictable traffic NoSQL	DynamoDB On-Demand	On-demand mode charges per request — no capacity planning needed for spiky workloads.



Analytics — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Process big data with Spark or Hadoop	EMR, Spark, Hadoop, big data cluster, Hive, Presto, managed Hadoop, map reduce	Amazon EMR	EMR is a managed cluster platform for big data frameworks like Spark and Hadoop.
Move data into Redshift	Redshift Spectrum, COPY command, data ingestion Redshift, load data	Redshift COPY / Kinesis Firehose	COPY is the fastest way to load data into Redshift; Firehose can auto-deliver streaming data.

warehouse, Firehose to Redshift

Deliver streaming data to S3 or Redshift

Firehose, Kinesis Data Firehose, delivery stream, load streaming data, buffer and deliver

Amazon Kinesis Data Firehose

Firehose buffers and delivers streaming data to S3, Redshift, OpenSearch, or Splunk.

Analyze streaming data with SQL

Kinesis Data Analytics, SQL on streams, real-time analytics, sliding window, tumbling window

Amazon Kinesis Data Analytics

Lets you run SQL queries on streaming data in real time.

Share data across accounts without copying

AWS Data Exchange, data subscription, third-party data, data marketplace, subscribe to dataset

AWS Data Exchange

Data Exchange lets you subscribe to and use third-party data sets directly in AWS.

Create visual dashboards from AWS data

QuickSight, embed dashboard, SPICE, BI dashboard, business report, ML insights QuickSight

Amazon QuickSight

QuickSight can embed dashboards into apps and uses SPICE for fast in-memory queries.

Coordinate data pipelines

Data Pipeline, data workflow, scheduled ETL, pipeline activity, EMR activity, data movement

AWS Data Pipeline

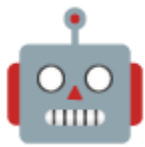
Data Pipeline orchestrates data movement and transformation between AWS services.



Application Integration — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Ensure messages are processed exactly once	exactly-once, FIFO queue, ordering, deduplication, SQS FIFO, message deduplication ID	Amazon SQS FIFO Queue	FIFO queues guarantee ordering and exactly-once processing — standard SQS does not.
Push mobile notifications	mobile push, APNs, FCM, GCM, push to device, SNS mobile, mobile endpoint	Amazon SNS (Mobile Push)	SNS integrates with APNs (Apple) and FCM (Google) for mobile push notifications.
Build visual workflows	Step Functions, visual workflow, state machine, human approval step, wait for callback	AWS Step Functions	Step Functions can pause a workflow and wait for a human approval token.
Connect SaaS apps to AWS	AppFlow, Salesforce to S3, ServiceNow, Slack data, SaaS integration, data flow SaaS	Amazon AppFlow	AppFlow transfers data between SaaS apps (Salesforce, Slack) and AWS services.
Send messages between applications reliably	MQ, ActiveMQ, RabbitMQ, message broker, open standard messaging, AMQP, MQTT, JMS	Amazon MQ	MQ is managed ActiveMQ/RabbitMQ for apps using standard messaging protocols.

Coordinate human approval in workflow	wait for token, human in the loop, approval task, callback, Step Functions task token	Step Functions (Task Token / Callback)	Task Token pattern lets a Step Functions state pause until a human or external system responds.
Expose internal microservice to other VPCs privately	PrivateLink, endpoint service, private service, share service across VPCs, NLB backed endpoint	AWS PrivateLink	PrivateLink exposes a service in your VPC to other VPCs privately through an endpoint.



Machine Learning — Extended

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Analyze sentiment of text	Comprehend, sentiment analysis, entity recognition, key phrase, NLP, text analysis, language detection	Amazon Comprehend	Comprehend extracts sentiment, entities, key phrases, and PII from text using NLP.
Forecast future values	Forecast, time-series forecasting, demand forecasting, predict future, sales forecast, inventory	Amazon Forecast	Forecast uses ML to produce accurate time-series forecasts without ML expertise.
Personalize recommendations	Personalize, recommendation engine, product recommendation,	Amazon Personalize	Personalize delivers real-time personalized recommendations —

	collaborative filtering, user behavior		same tech as Amazon.com.
Review code quality automatically	CodeGuru Reviewer, code review, pull request analysis, code quality, code defect, Java Python review	Amazon CodeGuru Reviewer	CodeGuru Reviewer uses ML to identify defects and suggest code improvements.
Profile and optimize application performance	CodeGuru Profiler, CPU bottleneck, heap usage, performance profiling, latency hotspot	Amazon CodeGuru Profiler	CodeGuru Profiler identifies the most expensive lines of code at runtime.
Generate text, images, or use foundation models	Bedrock, foundation model, generative AI, Claude on AWS, Titan, LLM, text generation, GenAI	Amazon Bedrock	Bedrock provides access to foundation models (Claude, Titan, Llama, etc.) via API.
Build AI agents that take actions	Bedrock Agents, agent, tool use, action group, orchestration, multi-step AI task	Amazon Bedrock Agents	Bedrock Agents can plan and execute multi-step tasks by calling APIs and tools.
Moderate user-generated content	Rekognition moderation, content moderation, unsafe image, explicit content, NSFW detection	Amazon Rekognition (Content Moderation)	Rekognition can automatically detect explicit or unsafe content in images and video.



Developer Tools & DevOps

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Store and version application code	CodeCommit, Git repository, source control, branch, pull request, managed Git	AWS CodeCommit	CodeCommit is AWS's managed private Git repository service.
Compile and test code automatically	CodeBuild, build server, compile, unit test, buildspec, build artifact, managed build	AWS CodeBuild	CodeBuild is a fully managed build service — no build servers to maintain.
Automate full release pipeline	CodePipeline, pipeline stages, source build deploy, release automation, CI/CD pipeline	AWS CodePipeline	CodePipeline orchestrates source → build → test → deploy stages automatically.
Deploy application to servers	CodeDeploy, deployment, rolling, blue/green, canary, appspec, deployment group	AWS CodeDeploy	CodeDeploy automates code deployments to EC2, Lambda, ECS, or on-prem servers.
Manage all developer tools in one place	CodeStar, project template, integrated IDE, developer portal, team collaboration	AWS CodeStar	CodeStar provides a unified UI to manage CodeCommit, CodeBuild, CodePipeline, and CodeDeploy.

Share libraries and packages across teams	CodeArtifact, package repository, npm, pip, Maven, artifact store, private package	AWS CodeArtifact	CodeArtifact is a managed artifact repository for npm, pip, Maven, and NuGet packages.
Automate OS patching on EC2 fleet	Patch Manager, patch baseline, patch group, SSM patch, patch compliance, patch window	AWS Systems Manager Patch Manager	Patch Manager automates OS and application patching across your EC2 fleet.
Run commands across many EC2 instances	Run Command, SSM Run Command, remote execution, fleet command, execute script	AWS Systems Manager Run Command	Run Command executes scripts or commands across multiple EC2 instances simultaneously.
Store operational data and parameters	Parameter Store, SSM Parameter Store, configuration data, SecureString, hierarchical parameter	AWS Systems Manager Parameter Store	Parameter Store stores config data and secrets — free tier available unlike Secrets Manager.
Track software inventory on EC2	SSM Inventory, software inventory, installed applications, instance metadata, fleet inventory	AWS Systems Manager Inventory	Inventory collects metadata about EC2 instances including installed software.



Architecture & Well-Architected

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Design for high availability	multiple AZ, redundancy, no single point of failure, failover, multi-region, HA design	Multi-AZ / Multi-Region Architecture	HA requires redundancy across AZs (same region) or regions (global DR).
Design for disaster recovery	RTO, RPO, backup restore, pilot light, warm standby, multi-site active-active, DR strategy	DR Strategies (Backup/Pilot Light/Warm/Active-Active)	RTO/RPO define the strategy — shorter = more expensive (active-active vs backup-restore).
Reduce latency for global users	CloudFront, Global Accelerator, edge caching, latency-based routing, global distribution	CloudFront / Global Accelerator / Route 53	Combine CDN, routing, and edge compute to serve global users with minimal latency.
Decouple architecture components	loose coupling, SQS between tiers, async, event-driven, decouple, independent scaling	SQS / SNS / EventBridge	Decoupling uses queues and events so components scale and fail independently.
Scale reads for a database	read replica, caching, ElastiCache, DAX,	Read Replicas + ElastiCache	Read replicas scale DB reads; ElastiCache caches frequent queries to reduce DB load.

	offload reads, reporting replica		
Review architecture best practices	Well-Architected, pillars, operational excellence, security, reliability, performance efficiency, cost optimization, sustainability	AWS Well-Architected Framework	The 6 pillars review is a Well-Architected Framework clue.
Run architecture review	Well-Architected Tool, WA Tool, workload review, risk, improvement plan	AWS Well-Architected Tool	The WA Tool is a self-service console tool for reviewing workloads against the framework.
Migrate workloads to AWS	Migration Hub, 7 Rs, rehost, replatform, refactor, retire, retain, relocate	AWS Migration Hub / 7 Rs	The 7 Rs (rehost/replatform/refactor etc.) framework guides migration strategy decisions.
Assess migration readiness	Migration Evaluator, TCO, business case, migration assessment, on-prem inventory	AWS Migration Evaluator	Migration Evaluator builds a business case by analyzing on-prem workloads and costs.
Move servers to AWS at scale	Application Migration Service, MGN, lift and shift, agentless replication, server migration	AWS Application Migration Service (MGN)	MGN replicates servers to AWS with minimal downtime — the primary lift-and-shift tool.



Global Infrastructure

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Choose a region	data residency, latency to users, compliance, regional service availability, region selection	AWS Region	Pick region based on: latency to users, data residency laws, service availability, cost.
Survive an Availability Zone failure	AZ failure, single AZ risk, spread across AZs, AZ redundancy, multi-AZ deployment	Multiple Availability Zones	AZs are isolated data centers in a region — distributing across AZs protects against AZ failure.
Run workloads close to users physically	edge location, Point of Presence, PoP, CloudFront PoP, 400+ locations	CloudFront Edge Locations / PoPs	CloudFront has 400+ edge locations globally — much more than the number of regions.
Understand AWS physical infrastructure	region, AZ, edge location, Local Zone, Outpost, data center	AWS Global Infrastructure	Know the hierarchy: Regions > AZs > Edge Locations / Local Zones > Outposts.



Compute — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Choose the right EC2 instance family	general purpose, compute optimized, memory optimized, storage optimized, accelerated computing, GPU	EC2 Instance Families	T/M = general; C = compute; R/X = memory; I/D = storage; P/G = GPU/ML.
Run GPU workloads on EC2	GPU instance, P4, G5, graphics, machine learning training, CUDA, deep learning, rendering	EC2 P/G Instance Types	P-family for ML training; G-family for graphics and inference. Both have GPU hardware.
Use burstable CPU performance	T3, T3a, T2, CPU credits, burstable, baseline CPU, credit balance, unlimited mode	EC2 T-Series (Burstable)	T instances earn CPU credits at baseline and spend them during bursts — good for variable workloads.
Launch instances in a placement group	cluster placement, spread placement, partition placement, low latency between instances, HPC	EC2 Placement Groups	Cluster = low latency; Spread = max resilience; Partition = large distributed systems.
Protect instance from accidental termination	termination protection, disable termination, instance protection, stop protection	EC2 Termination Protection	Enable termination protection to prevent accidental stop/terminate via console or API.
Retrieve instance	instance metadata, 169.254.169.254, IMDSv2,	EC2 Instance Metadata Service (IMDS)	Applications running on EC2 query the metadata endpoint at

metadata from inside EC2	instance identity, metadata endpoint		169.254.169.254 for instance info.
Run containers on EC2 without Kubernetes	ECS, task definition, ECS service, task, Elastic Container Service, Docker on AWS	Amazon ECS	ECS runs Docker containers on EC2 or Fargate — simpler than EKS if Kubernetes isn't needed.
Run containers with Kubernetes	EKS, Kubernetes, kubectl, pod, node group, managed node group, K8s	Amazon EKS	EKS is managed Kubernetes — use when you need Kubernetes-specific features or portability.
Store container images privately	ECR, Elastic Container Registry, Docker image, container image, push image, pull image	Amazon ECR	ECR is the private Docker registry tightly integrated with ECS and EKS.
Run a function in response to S3 upload	S3 trigger, S3 event, Lambda trigger, object created, put event, Lambda on upload	Lambda (S3 Event Trigger)	S3 can invoke Lambda directly on object creation — classic event-driven pattern.
Set max execution time for a Lambda	Lambda timeout, function timeout, max duration, 15 minutes, execution limit	Lambda Timeout	Lambda max timeout is 15 minutes. For longer jobs use Step Functions, Fargate, or Batch.
Run Lambda in your VPC	Lambda VPC, Lambda private subnet, Lambda access RDS, VPC Lambda, ENI	Lambda VPC Configuration	Lambda can be placed inside a VPC to access private resources like

			RDS — adds cold start latency.
Reuse code across Lambda functions	Lambda Layer, shared library, dependency layer, common code, layer ARN	Lambda Layers	Layers let multiple functions share libraries, reducing package size and duplication.
Handle millions of invocations per second	Lambda concurrency, reserved concurrency, provisioned concurrency, throttle, burst limit	Lambda Concurrency	Reserved concurrency caps a function's scale; provisioned keeps instances warm for low latency.



Storage — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Choose the right S3 storage class	S3 Standard, S3-IA, S3 One Zone-IA, Intelligent-Tiering, Glacier, Deep Archive, storage cost	S3 Storage Classes	Standard → frequent access; IA → infrequent; Glacier → archive; Deep Archive → cheapest long-term.
Make S3 object publicly accessible	bucket policy, public access, ACL, block public access, presigned URL, public read	S3 Bucket Policy / Presigned URL	Block Public Access is on by default; presigned URLs grant temporary access without making bucket public.

Host a static website on S3	static website hosting, index.html, error.html, S3 website endpoint, bucket website	S3 Static Website Hosting	Enable static website hosting on a bucket and optionally front it with CloudFront.
Encrypt objects stored in S3	SSE-S3, SSE-KMS, SSE-C, server-side encryption, client-side encryption, S3 encryption	S3 Encryption Options	SSE-S3 = AWS managed keys; SSE-KMS = KMS keys (auditable); SSE-C = customer-provided keys.
Control access to S3 from specific role	bucket policy, IAM policy, S3 ACL, resource-based policy, deny, allow, condition key	S3 Bucket Policy vs IAM Policy	Bucket policies are resource-based; IAM policies are identity-based. Both can grant/deny S3 access.
Audit who accessed S3 objects	S3 server access logging, S3 access logs, request logging, who accessed, S3 audit	S3 Server Access Logging	S3 access logs capture every request to a bucket — stored in another S3 bucket.
Reduce EBS volume costs for infrequently used data	EBS cold HDD, sc1, throughput optimized HDD, st1, magnetic, archive workload	EBS sc1 / st1 Volume Types	sc1 = cold HDD for infrequent; st1 = throughput HDD for sequential big data; gp3 for general.
Take automated snapshots of EBS	DLM, Data Lifecycle Manager, EBS snapshot policy, automated snapshot, snapshot schedule	Amazon DLM (Data Lifecycle Manager)	DLM automates creation, retention, and deletion of EBS snapshots on a schedule.

Share EBS snapshots across accounts	snapshot share, cross-account snapshot, AMI copy, share snapshot	EBS Snapshot Sharing	EBS snapshots can be shared with specific AWS accounts or made public.
Access S3 from EC2 without internet	VPC endpoint, gateway endpoint, S3 VPC endpoint, private S3 access, no NAT	S3 Gateway VPC Endpoint	S3 and DynamoDB support gateway endpoints — free and keep traffic on AWS backbone.



Database — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Encrypt an existing RDS database	encrypt RDS, encryption at rest, KMS RDS, snapshot copy encrypt, enable encryption	RDS Encryption (via Snapshot)	You can't enable encryption on a running RDS instance — snapshot it, copy with encryption, restore.
Connect to RDS without storing credentials in code	Secrets Manager RDS rotation, IAM database authentication, rotate credentials, no hardcoded password	Secrets Manager / IAM DB Authentication	Secrets Manager rotates RDS passwords automatically; IAM DB auth uses tokens instead of passwords.
Audit all database queries	RDS audit log, database activity streams, CloudWatch RDS logs, query log, general log	RDS Logs / Database Activity Streams	Enable general/audit logs to CloudWatch; Activity Streams provide

			near-real-time audit for Aurora.
Scale DynamoDB throughput instantly	DynamoDB auto scaling, on-demand mode, provisioned throughput, RCU, WCU, capacity mode	DynamoDB Capacity Modes	On-demand scales instantly with no planning; provisioned is cheaper for predictable workloads.
Model hierarchical or relational data in DynamoDB	single table design, GSI, LSI, composite key, sort key, adjacency list, DynamoDB modeling	DynamoDB Advanced Modeling	Single-table design with GSIs models relationships in DynamoDB — different from SQL thinking.
Back up DynamoDB with point-in-time recovery	PITR, DynamoDB backup, point-in-time, continuous backup, restore to second	DynamoDB PITR	PITR lets you restore a DynamoDB table to any second within the last 35 days.
Export DynamoDB data to S3	DynamoDB export, export to S3, no-impact export, S3 bucket export, full export	DynamoDB Export to S3	DynamoDB can export table data to S3 in JSON or Parquet format without affecting performance.
Run SQL queries without managing a database server	Aurora Serverless, RDS Proxy, serverless relational, auto pause, Data API	Aurora Serverless v2	Aurora Serverless v2 scales instantly and fine-grained — no server management needed.
Reduce connection overhead to RDS	RDS Proxy, connection pooling, Lambda to RDS, too many connections, proxy	Amazon RDS Proxy	RDS Proxy pools connections between Lambda/apps and RDS

— essential for Lambda
→ RDS patterns.

Replicate data from on-prem database to AWS

DMS replication, CDC, change data capture, ongoing replication, live migration

AWS DMS (Ongoing Replication)

DMS supports full-load + CDC for live ongoing replication with minimal downtime.



Networking — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Assign a static public IP to an EC2 instance	Elastic IP, EIP, static IP, persistent IP, reassign IP, fixed public IP	Elastic IP Address (EIP)	EIPs are static IPv4 addresses you own — they persist across instance stop/start.
Allow EC2 instances to communicate privately	private IP, same VPC, internal communication, VPC routing, private subnet traffic	VPC Internal Routing	Resources in the same VPC communicate via private IPs using VPC's local route.
Filter traffic between subnets in same VPC	security group, NACL, east-west traffic, micro-segmentation, intra-VPC filtering	Security Groups / NACLs	Security Groups are stateful per-instance; NACLs are stateless per-subnet.

Capture network traffic for analysis	VPC Flow Logs, network traffic, packet capture, IP traffic, flow record, Athena query flows	VPC Flow Logs	Flow Logs capture metadata about IP traffic in your VPC — not packet content, just headers.
Mirror actual network packets	Traffic Mirroring, packet capture, deep packet inspection, intrusion detection, network forensics	VPC Traffic Mirroring	Traffic Mirroring copies actual packet data (not just flow metadata) to a monitoring appliance.
Access AWS services without internet gateway	interface endpoint, PrivateLink endpoint, VPC endpoint, private DNS, AWS service private access	Interface VPC Endpoint (PrivateLink)	Interface endpoints use ENIs with private IPs — supports most AWS services unlike gateway endpoints.
Advertise on-prem routes to AWS via Direct Connect	BGP, Border Gateway Protocol, virtual interface, private VIF, public VIF, transit VIF	Direct Connect Virtual Interfaces	Direct Connect uses BGP — private VIF for VPC access, public VIF for AWS public services.
Add redundancy to Direct Connect	Direct Connect backup VPN, redundant Direct Connect, failover connection, resiliency	Direct Connect + VPN Backup	Best practice: use a VPN as failover for Direct Connect to ensure hybrid connectivity.
Assign IPv6 to VPC resources	IPv6, dual-stack, IPv6 CIDR, egress-only IGW, IPv6 subnet	VPC IPv6 / Egress-Only Internet Gateway	Egress-only IGW allows IPv6 outbound traffic from private subnets (like NAT Gateway for IPv4).

Simplify network management at scale

AWS Network Manager, global network, SD-WAN, WAN monitoring, Transit Gateway network

AWS Network Manager

Network Manager provides a central console to manage and monitor your global network.



Identity & Security — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Evaluate IAM policy permissions	IAM policy simulator, test policy, evaluate permissions, policy evaluation, access denied debug	IAM Policy Simulator	The Policy Simulator lets you test IAM policies to see if an action is allowed or denied.
Understand why access was denied	access analyzer, policy finding, unused access, external access, resource exposure	IAM Access Analyzer	Access Analyzer identifies resources exposed to external principals and unused permissions.
Apply least privilege automatically	IAM Access Advisor, last accessed, unused permissions, prune permissions, service last used	IAM Access Advisor	Access Advisor shows the last time each service was accessed — helps prune excess permissions.
Prevent privilege escalation	permission boundary, max permissions,	IAM Permission Boundaries	Permission boundaries cap the maximum permissions an identity

	iam:PassRole, privilege escalation prevention		can have, even if policies allow more.
Detect publicly exposed resources	Access Analyzer, public S3, public SQS, internet-facing, resource-based policy exposure	IAM Access Analyzer	Access Analyzer flags resources accessible outside your account — S3, KMS, Lambda, SQS, etc.
Sign API requests to AWS	SigV4, Signature Version 4, request signing, HMAC, signed request, AWS SDK signing	AWS Signature Version 4 (SigV4)	All AWS API calls must be signed with SigV4 — the SDK does this automatically.
Manage certificates on-premises or other CAs	Private CA, ACM PCA, private certificate authority, internal PKI, private cert	AWS Private CA (ACM PCA)	ACM Private CA creates a managed private CA for issuing internal TLS certificates.
Block specific countries from your application	WAF geo match, geographic restriction, country block, WAF rule, geo rule	AWS WAF Geo Match Rule	WAF Geo Match rules allow or block requests based on the country of origin.
Detect exposed credentials in code	secret detection, GitLeaks, credentials in code, Macie, CodeGuru security finding	Amazon CodeGuru Security / Macie	CodeGuru scans code for hardcoded secrets; Macie scans S3 for exposed credentials.
Implement zero-trust network access	zero trust, ZTNA, verified access, identity-based access, no VPN, application-level access	AWS Verified Access	Verified Access provides secure access to applications without a VPN, using identity context.



Monitoring & Logging — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Set up a billing alarm	billing alarm, cost alarm, CloudWatch billing, estimated charges, free tier alarm	CloudWatch Billing Alarm	Create a CloudWatch alarm on the EstimatedCharges metric — must be in us-east-1.
Monitor Lambda function errors	Lambda metrics, invocation errors, throttles, duration, concurrent executions, error rate	CloudWatch Lambda Metrics	Lambda publishes Invocations, Errors, Duration, and Throttles metrics to CloudWatch automatically.
Stream CloudWatch logs to third-party tools	Kinesis, log subscription, Splunk, Datadog, log forwarding, Firehose subscription filter	CloudWatch Logs Subscription	Subscription filters stream log data to Kinesis, Firehose, or Lambda in real time.
Get notified of CloudTrail events	CloudTrail + EventBridge, API event alert, alert on API call, root login alert	CloudTrail + EventBridge	EventBridge can trigger on specific CloudTrail events — e.g., alert when root user logs in.
Retain audit logs immutably	CloudTrail log file integrity, log file validation, immutable audit, tamper detection	CloudTrail Log File Integrity	CloudTrail can validate log file integrity using SHA-256 hashing to detect tampering.

Monitor ECS container metrics	Container Insights, ECS metrics, task CPU memory, container performance, ECS monitoring	CloudWatch Container Insights	Container Insights collects metrics and logs from ECS, EKS, and K8s clusters.
Create a dashboard visible to the team	CloudWatch Dashboard, shared dashboard, cross-account dashboard, metric widget, alarm widget	CloudWatch Dashboards	Dashboards are customizable pages of widgets showing metrics and alarms across services.
Track application-level custom events	custom metric, PutMetricData, EMF, Embedded Metric Format, application telemetry	CloudWatch Custom Metrics / EMF	You can push any metric using PutMetricData; EMF embeds metrics in structured logs.
Detect account root login	CloudTrail, root user activity, root login alert, EventBridge rule, SNS alert root	CloudTrail + EventBridge + SNS	Classic pattern: CloudTrail → EventBridge rule for root user activity → SNS email notification.



Application Integration — Deep Dive

If the question says...

Look for these keywords

Answer is...

Why this is the clue

Delay message processing	SQS delay queue, message timer, DelaySeconds, visibility timeout, deferred processing	SQS Delay Queues	Delay queues postpone delivery of new messages for up to 15 minutes.
Handle failed SQS messages	dead-letter queue, DLQ, failed message, maxReceiveCount, redrive policy, poison pill	SQS Dead-Letter Queue (DLQ)	After maxReceiveCount failures, messages go to the DLQ for inspection and reprocessing.
Temporarily hide a message being processed	visibility timeout, message in-flight, processing time, extend timeout, ChangeMessageVisibility	SQS Visibility Timeout	Visibility timeout hides a message from other consumers while one consumer processes it.
Poll SQS efficiently and reduce costs	long polling, WaitTimeSeconds, reduce empty responses, efficient polling, 20 seconds	SQS Long Polling	Long polling waits up to 20s for a message before returning — reduces empty responses and cost.
Pass large payloads through SQS	SQS extended client, S3 pointer, large message, 256 KB limit, message size limit	SQS Extended Client Library	SQS max message size is 256 KB; the extended client stores the payload in S3 and sends a pointer.
Order API calls without code orchestration	API Gateway, request validation, mapping template, integration, method response, mock integration	Amazon API Gateway	API Gateway handles auth, throttling, request/response transformation without custom code.

Protect API from too many requests	API Gateway throttling, rate limit, usage plan, API key, burst limit, quota	API Gateway Throttling / Usage Plans	Usage Plans limit requests per second and per month; throttling protects backend services.
Cache API responses	API Gateway cache, response cache, reduce backend calls, cache TTL, stage cache	API Gateway Response Caching	Enable stage-level caching to return cached responses and reduce backend invocations.
Webhook from SaaS into AWS	EventBridge API destination, API destination, webhook, inbound event, partner event source	EventBridge API Destinations / Partner Sources	API Destinations send events to external HTTP endpoints; partner sources receive SaaS events.



Machine Learning — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Label training data	SageMaker Ground Truth, data labeling, human labeling, annotation, RLHF labels, labeling workforce	Amazon SageMaker Ground Truth	Ground Truth manages human labeling workflows with active learning to reduce labeling cost.

Track ML experiments	SageMaker Experiments, experiment tracking, model comparison, trial, run, hyperparameter comparison	Amazon SageMaker Experiments	Experiments tracks parameters, metrics, and artifacts across training runs for comparison.
Deploy ML model as an endpoint	SageMaker endpoint, real-time inference, model deployment, inference endpoint, A/B test models	SageMaker Real-Time Endpoints	Deploy models as HTTPS endpoints for real-time inference — supports A/B testing with variants.
Run batch inference on large datasets	SageMaker Batch Transform, offline inference, batch prediction, no endpoint, transform job	SageMaker Batch Transform	Batch Transform runs inference on large datasets without hosting a persistent endpoint.
Monitor model accuracy over time	SageMaker Model Monitor, data drift, model drift, baseline, monitoring schedule	SageMaker Model Monitor	Model Monitor detects data and model quality drift compared to training baseline.
Tune hyperparameters automatically	SageMaker Automatic Model Tuning, HPO, hyperparameter optimization, Bayesian optimization	SageMaker Automatic Model Tuning	Runs multiple training jobs with different hyperparameter combinations to find the best model.
Build ML pipelines	SageMaker Pipelines, ML workflow, pipeline step, training pipeline, CI/CD ML	SageMaker Pipelines	Pipelines orchestrates multi-step ML workflows from data prep to model deployment.

Store and share ML features	SageMaker Feature Store, feature group, feature reuse, training serving skew, online offline	SageMaker Feature Store	Feature Store provides a central repository for ML features — online for inference, offline for training.
Analyze structured data automatically	SageMaker Autopilot, AutoML, auto model selection, no-code ML, tabular data, automatic training	SageMaker Autopilot	Autopilot automatically selects algorithms, trains, and tunes models on tabular data.
Add recommendations without ML expertise	Personalize, collaborative filtering, user-item interaction, recommendation API, real-time recs	Amazon Personalize	Personalize handles all ML complexity — you provide interactions data and get recommendation APIs.
Detect fraud in transactions	Fraud Detector, fraud model, transaction scoring, account takeover, fraud ring, risk score	Amazon Fraud Detector	Fraud Detector uses ML trained on Amazon's own fraud data to score transaction risk.
Extract medical information from text	Comprehend Medical, clinical notes, ICD-10, RxNorm, medical entity, PHI detection	Amazon Comprehend Medical	Comprehend Medical extracts medical entities (diagnoses, medications, dosages) from clinical text.



Architecture & Well-Architected — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Design lowest possible RTO	active-active, multi-region active, real-time replication, zero downtime, instant failover	Multi-Region Active-Active	Active-active is the most expensive but gives near-zero RTO. Other strategies have higher RTO.
Design lowest possible RPO	synchronous replication, no data loss, RDS synchronous standby, Aurora global, zero RPO	Synchronous Replication (Multi-AZ / Aurora Global)	Synchronous replication (RDS Multi-AZ, Aurora) gives near-zero RPO — data is always in sync.
Reduce cost of disaster recovery	pilot light, minimal standby, warm standby, cold standby, backup and restore, cheapest DR	Pilot Light / Backup & Restore	Backup-Restore is cheapest (highest RTO/RPO); pilot light keeps minimal resources warm.
Decouple a monolithic application	strangler fig, microservices, decouple, SQS between tiers, event-driven, API gateway front	Microservices + SQS/SNS + API Gateway	Break monolith into services communicating via queues/events — API Gateway fronts HTTP traffic.
Handle traffic spikes without pre-scaling	serverless, Lambda, DynamoDB on-demand, SQS buffer, auto scaling, reactive scaling	Serverless / On-Demand Services	Serverless services scale instantly — no pre-warming. SQS buffers spikes before backend processes them.

Ensure application can handle AZ failure	multi-AZ deployment, spread instances, AZ-aware, ELB across AZs, RDS Multi-AZ	Multi-AZ with ELB + Auto Scaling	ELB distributes across AZs; Auto Scaling replaces failed instances; RDS Multi-AZ fails over automatically.
Reduce latency for database reads	ElastiCache, read replica, DAX, caching strategy, lazy loading, write-through, cache-aside	ElastiCache / DAX / Read Replicas	Cache frequently read data in ElastiCache; DAX for DynamoDB; read replicas offload heavy reads.
Design for data sovereignty	data residency, GDPR, compliance, keep data in region, regional storage, data localization	Region Selection + S3 / RDS in specific region	AWS data stays in the region where you create it unless you explicitly replicate it.
Make application resilient to cascading failures	circuit breaker, bulkhead, retry with backoff, timeout, resilience pattern	Resilience Patterns (Circuit Breaker / Bulkhead)	Use exponential backoff, circuit breakers, and bulkheads to prevent cascade failures.
Share infrastructure templates across teams	CloudFormation StackSets, Service Catalog, template sharing, multi-account deploy, guardrails	CloudFormation StackSets / AWS Service Catalog	StackSets deploy CloudFormation stacks across accounts; Service Catalog provides approved templates.
Review costs as part of architecture	Cost pillar, Well-Architected cost review, right-sizing, elasticity, consumption model	Well-Architected Cost Optimization Pillar	Cost Optimization pillar: use managed services, right-size, match capacity to demand, use spot.



Developer Tools & DevOps — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Automatically test code changes	CodeBuild test, unit tests, integration tests, test report, test spec, build test phase	AWS CodeBuild	CodeBuild can run automated tests as part of the build phase before deployment.
Roll back a failed deployment	CodeDeploy rollback, automatic rollback, deployment failure, previous revision, rollback trigger	CodeDeploy Automatic Rollback	CodeDeploy can automatically roll back if a deployment fails or alarms are triggered.
Deploy to production with approval gate	CodePipeline manual approval, approval action, human approval, gate, approve before deploy	CodePipeline Manual Approval Action	Add a Manual Approval action in CodePipeline to require human sign-off before production deploy.
Enforce consistent infrastructure across accounts	CloudFormation StackSets, multi-account, multi-region, organizational deployment, drift detection	CloudFormation StackSets	StackSets deploy one CloudFormation template across hundreds of accounts simultaneously.

Detect infrastructure drift	CloudFormation drift detection, stack drift, resource drift, configuration change outside IaC	CloudFormation Drift Detection	Drift detection compares actual resource configuration to the CloudFormation template.
Import existing resources into CloudFormation	CloudFormation import, resource import, existing resource, bring under management	CloudFormation Resource Import	You can import existing AWS resources into a CloudFormation stack without recreating them.
Preview CloudFormation changes before applying	change set, CloudFormation change set, what will change, preview changes, plan	CloudFormation Change Sets	Change Sets show exactly what CloudFormation will add, modify, or delete before execution.
Manage configuration across environments	AppConfig, feature flag, configuration deployment, gradual rollout, configuration management	AWS AppConfig	AppConfig deploys application configuration changes safely with validation and rollback.
Toggle features without redeploying	feature flag, AppConfig, LaunchDarkly integration, feature toggle, dark launch	AWS AppConfig (Feature Flags)	AppConfig can manage feature flags — enable/disable features without a code deployment.
Trace slow API calls end-to-end	X-Ray trace, service map, segment, subsegment, latency breakdown, trace ID, cold start trace	AWS X-Ray	X-Ray provides end-to-end distributed tracing — identifies latency bottlenecks across services.



Global Infrastructure — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Run low-latency workloads in a metro area	Local Zone, single-digit millisecond, city, metropolitan, local compute, local AWS	AWS Local Zones	Local Zones extend AWS into major metros — ideal for media rendering, gaming, real-time apps.
Extend AWS into your own facility	Outposts, on-prem AWS, Outposts rack, Outposts server, local AWS hardware, data center	AWS Outposts	Outposts brings native AWS services and APIs into your own data center or colocation facility.
Use AWS at the network edge	Wavelength, 5G edge, mobile network, ultra-low latency mobile, telecom edge, Verizon AWS	AWS Wavelength	Wavelength embeds AWS compute inside 5G networks — single-digit millisecond for mobile apps.
Understand number of regions vs AZs vs edge	30+ regions, 90+ AZs, 400+ edge locations, points of presence, global footprint	AWS Global Infrastructure Numbers	Regions (~33+), AZs (multiple per region, ~3-6), Edge Locations (400+). Edge >> AZ >> Region count.
Choose between single-region and multi-region	regional vs global, data residency, latency, cost, complexity, DR	Single vs Multi-Region Design	Single-region: simpler and cheaper. Multi-region: needed for <1min RTO, global users, or compliance.

Deploy to government or regulated workloads	GovCloud, FedRAMP, ITAR, regulated, government cloud, us-gov-east, us-gov-west	AWS GovCloud (US)	GovCloud regions are isolated for US government agencies and contractors with ITAR compliance.
Ensure data does not leave a country	data sovereignty, data residency guarantee, keep in country, GDPR data boundary	AWS Region + Data Residency Controls	AWS does not move your data out of a region without your explicit action (e.g., enabling replication).



Migration & Transfer

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Migrate virtual machines to AWS	Application Migration Service, MGN, server migration, lift and shift, rehost, agentless	AWS Application Migration Service (MGN)	MGN (previously SMS) is the primary tool for lifting and shifting VMs to EC2.
Assess on-prem environment before migration	Migration Evaluator, TCO calculator, inventory collection, business case, discovery	AWS Migration Evaluator	Migration Evaluator collects on-prem data and produces a cost-justified migration business case.
Discover on-prem servers automatically	Application Discovery Service, ADS, agent-based discovery,	AWS Application Discovery Service	ADS discovers on-prem servers, dependencies,

	agentless discovery, inventory, dependencies		and performance data to plan migrations.
Track migration progress centrally	Migration Hub, migration tracker, portfolio, migration status, server tracking	AWS Migration Hub	Migration Hub gives a single pane of glass to track migrations across tools and AWS services.
Migrate a database with minimal downtime	DMS, database migration, CDC, change data capture, live migration, cut-over	AWS Database Migration Service (DMS)	DMS migrates databases continuously using CDC so you can cut over with seconds of downtime.
Transfer terabytes of data online	DataSync, online transfer, scheduled sync, accelerated transfer, NFS to S3, SMB to S3	AWS DataSync	DataSync accelerates online data transfers — faster than cp/rsync and handles scheduling.
Transfer petabytes offline	Snowball Edge, Snowmobile, physical transfer, offline migration, ship device, bandwidth limited	AWS Snowball / Snowmobile	Use Snowball when network transfer would take too long; Snowmobile for exabyte-scale.
Replatform databases to Aurora	schema conversion, SCT, heterogeneous, Oracle to Aurora, SQL Server to MySQL, convert schema	AWS Schema Conversion Tool (SCT)	SCT converts database schema and stored procedures from one engine to another.
Migrate mainframe workloads	mainframe migration, COBOL, z/OS, Micro Focus, AWS mainframe	AWS Mainframe Modernization	Provides tools and managed runtime to migrate and modernize

modernization, refactor
mainframe

COBOL/PL1 mainframe
workloads.



Compliance & Governance

**If the question
says...**

**Look for these
keywords**

Answer is...

Why this is the clue

Meet PCI DSS
compliance
requirements

PCI, cardholder data,
payment, compliance
report, PCI DSS, audit
report

**AWS Artifact +
PCI DSS Shared
Responsibility**

AWS Artifact provides PCI
compliance reports;
workloads must also follow
customer PCI obligations.

Access AWS
compliance
reports

AWS Artifact, SOC
report, ISO
certification,
compliance
document, audit
report, attestation

AWS Artifact

Artifact is the self-service
portal for downloading AWS
compliance reports and
agreements.

Enforce
governance
policies across
accounts

SCP, service control
policy, Organizations,
preventive guardrail,
deny action, guardrail

**AWS
Organizations
SCPs**

SCPs are preventive
guardrails — they restrict
what member accounts can
do regardless of IAM policies.

Detect
non-compliant
resources
automatically

Config rule, detective
guardrail,
non-compliant,
auto-remediation,
conformance pack

**AWS Config
Rules**

Config Rules continuously
evaluate resources and flag
non-compliance — can
auto-remediate.

Audit who did what and when	CloudTrail, audit log, API history, governance, who called API, accountability	AWS CloudTrail	CloudTrail records all API activity — the definitive audit log for AWS account actions.
Implement data classification	Macie, data classification, sensitive data, PII, PHI, S3 classification, data privacy	Amazon Macie	Macie automatically classifies data in S3 buckets and alerts on sensitive data exposure.
Apply consistent tags for governance	Tag policies, Organizations tag policy, required tags, tag compliance, enforce tagging	AWS Organizations Tag Policies	Tag Policies enforce consistent tagging across all accounts in an organization.
Control what regions accounts can use	SCP, deny region, restrict region, allowed regions, organizational policy	SCP Region Restriction	Add a Deny SCP with a condition on aws:RequestedRegion to block usage of specific regions.
Review security posture holistically	Security Hub, score, security standard, CIS, FSBP, AWS Foundational Security Best Practices	AWS Security Hub	Security Hub aggregates security findings and scores your posture against security standards.



Exam Traps & Common Confusions

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Confused between CloudTrail vs CloudWatch	API audit vs metrics/alarms, who called API vs CPU alarm	CloudTrail = API audit log; CloudWatch = metrics & alarms	CloudTrail records <i>who did what</i> . CloudWatch monitors <i>operational health</i> . Never mix them up.
Confused between Security Group vs NACL	stateful vs stateless, instance vs subnet, allow only vs allow/deny	SG = stateful, instance-level, allow only; NACL = stateless, subnet-level, allow + deny	SGs remember return traffic automatically. NACLs need explicit inbound AND outbound rules.
Confused between S3 vs EBS vs EFS	object store vs block vs file, EC2 attached vs shared vs web	S3 = object/web; EBS = block/single EC2; EFS = shared file/NFS	EBS can only attach to one EC2 at a time (multi-attach is a special mode). EFS mounts to many.
Confused between RDS Read Replica vs Multi-AZ	read scaling vs HA, async vs sync, reporting vs failover	Read Replica = async, read scaling; Multi-AZ = sync, automatic failover	Read Replicas improve performance. Multi-AZ improves availability. They serve different purposes.
Confused between Secrets Manager vs Parameter Store	rotation vs static config, cost vs free	Secrets Manager = auto-rotation, costs money; Parameter Store = free, no auto-rotation	Secrets Manager rotates automatically (great for DB passwords). Parameter Store is free for standard params.

Confused between Direct Connect vs VPN	dedicated line vs internet tunnel, consistent bandwidth vs variable	Direct Connect = private dedicated bandwidth; VPN = encrypted over public internet	Direct Connect never touches the internet. VPN does but encrypts traffic with IPsec.
Confused between SNS vs SQS	push vs pull, fan-out vs queue, pub-sub vs decoupling	SNS = push/fan-out to many; SQS = pull/buffer between two services	SNS pushes to all subscribers simultaneously. SQS holds messages until a consumer polls.
Confused between Kinesis vs SQS	streaming vs queuing, ordered vs unordered, replay vs delete	Kinesis = ordered streaming, replayable; SQS = unordered queue, message deleted after consume	Kinesis retains data for replay (default 24h). SQS deletes messages after successful consumption.
Confused between CloudFront vs Global Accelerator	HTTP/HTTPS caching vs TCP/UDP routing, CDN vs network routing	CloudFront = CDN with caching; Global Accelerator = network-layer routing, no caching	CloudFront caches content at edge. Global Accelerator routes TCP/UDP traffic over AWS backbone without caching.
Confused between ECS vs EKS vs Fargate	AWS-native vs Kubernetes vs serverless containers	ECS = AWS-native; EKS = Kubernetes; Fargate = serverless runtime for ECS or EKS	Fargate is a <i>launch type</i> , not a standalone service. ECS/EKS can both use Fargate as the compute.
Confused between IAM User vs IAM Role	permanent credentials vs temporary, human vs service	User = permanent credentials for humans; Role = temporary credentials	Roles use STS to issue short-lived tokens. Never embed IAM user

		for services or cross-account	keys in EC2 — use an instance role.
Confused between Elastic Beanstalk vs CloudFormation	PaaS vs IaC, app deployment vs infra provisioning	Beanstalk = deploy your app, AWS manages infra; CloudFormation = you define all infra as code	Beanstalk is opinionated and easier. CloudFormation is flexible and covers any AWS resource.
Confused between Pilot Light vs Warm Standby	minimal vs partial, cold resources vs scaled-down running	Pilot Light = core services running, rest off; Warm Standby = fully running but smaller scale	Pilot Light needs time to scale up on failover. Warm Standby can handle traffic immediately at reduced capacity.
Confused between aws:SourceIp vs aws:VpcSourceIp	public IP condition vs private IP condition, internet vs VPC	aws:SourceIp = public/internet IP; aws:VpcSourceIp = private IP within VPC	Use aws:VpcSourceIp when traffic comes through a VPC endpoint — aws:SourceIp won't work there.
Confused between S3 Transfer Acceleration vs DataSync	fast S3 upload via edge vs scheduled sync from on-prem	Transfer Acceleration = fast S3 uploads using CloudFront edges; DataSync = scheduled on-prem sync	Transfer Acceleration speeds up direct S3 uploads globally. DataSync syncs NFS/SMB shares to AWS.



Scenario-Based Patterns (Multi-Service Answers)

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Serverless web application	no server, API, frontend, Lambda, scalable web app, event-driven app	S3 (static) + CloudFront + API Gateway + Lambda + DynamoDB	Classic serverless stack: static site on S3, API via Gateway + Lambda, data in DynamoDB.
Highly available 3-tier web app	web tier, app tier, DB tier, multi-AZ, auto scaling, HA	Route 53 + ALB + EC2 Auto Scaling + RDS Multi-AZ	Standard 3-tier: Route 53 → ALB → Auto Scaling EC2 group → RDS Multi-AZ.
Protect API from DDoS and injection attacks	API protection, WAF, DDoS, injection, rate limit, CloudFront	CloudFront + AWS WAF + API Gateway + AWS Shield	Layer CloudFront for CDN, WAF for app-layer filtering, Shield for volumetric DDoS.
Process images uploaded by users	image upload, resize, thumbnail, Lambda trigger, S3 event, async processing	S3 + S3 Event → Lambda + S3 output	User uploads to S3 → S3 triggers Lambda → Lambda processes and saves result to another S3 bucket.
Send email notifications on events	email alert, SNS email, CloudWatch alarm notification, notify on event	CloudWatch Alarm / EventBridge → SNS → Email	SNS email endpoint sends notifications triggered by CloudWatch or EventBridge rules.

Real-time clickstream analytics	clickstream, web analytics, real-time, stream, dashboard	Kinesis Data Streams → Kinesis Data Analytics → Kinesis Firehose → S3/Redshift	Stream events through Kinesis, analyze in real time, deliver to S3 or Redshift for BI.
Centralize logs from hundreds of accounts	cross-account logs, centralized logging, organization logs, log archive	CloudWatch Logs → Kinesis Firehose → S3 (log archive account)	Use Firehose subscription filters to stream logs from many accounts into a central S3 bucket.
IoT sensor data to dashboard	IoT, sensor, device, telemetry, MQTT, dashboard, real-time	AWS IoT Core → Kinesis → Lambda/Timestream → QuickSight	IoT Core ingests MQTT, Kinesis streams data, Timestream stores time-series, QuickSight visualizes.
Machine learning pipeline	data prep, train model, deploy endpoint, monitor, retrain, ML ops	S3 + Glue → SageMaker Pipelines → SageMaker Endpoint + Model Monitor	S3 stores data, Glue preps it, SageMaker Pipelines trains/deploys, Model Monitor watches drift.
Hybrid storage with on-prem access	on-prem access to cloud storage, file gateway, NFS to cloud, local cache	AWS Storage Gateway (File Gateway) → S3	File Gateway presents S3 as an NFS share to on-prem applications with local caching.
Cross-region disaster recovery	DR, failover, another region, backup region, route traffic	Route 53 health checks + failover routing → secondary region	Route 53 failover routing detects unhealthy primary and routes to secondary region automatically.

Audit all changes to production AWS resources	change audit, who changed, configuration drift, compliance, history	CloudTrail + AWS Config + Config Rules	CloudTrail logs <i>who</i> made API calls; Config records <i>what</i> changed and evaluates compliance.
Reduce database load with caching	read-heavy, slow queries, cache, reduce DB calls, session store	ElastiCache (Redis/Memcached) in front of RDS	Cache frequent reads in ElastiCache. Redis also handles session storage, leaderboards, pub/sub.
Securely expose internal microservice to other teams	internal API, private service, no internet, cross-account service	NLB + AWS PrivateLink (VPC Endpoint Service)	PrivateLink exposes an NLB-backed service privately to other VPCs without peering or internet.
Run nightly batch ETL jobs cost-effectively	batch, nightly, ETL, Spark, cost-effective, scheduled, transient cluster	EventBridge Scheduler → AWS Glue Job or EMR (transient cluster)	Trigger Glue on a schedule for serverless ETL; use transient EMR for Spark — terminate after job.



Limits, Quotas & Numbers Worth Knowing

If the question says...

Look for these keywords

Answer is...

Why this is the clue

Maximum Lambda execution time	Lambda timeout, max duration, long running Lambda, 15 minutes	15 minutes	Lambda hard limit is 900 seconds (15 min). Use Step Functions + Lambda or Fargate for longer jobs.
Maximum SQS message size	SQS payload, message size, large message, 256 KB	256 KB	SQS max is 256 KB per message. Use S3 + pointer pattern (SQS Extended Client) for larger payloads.
Maximum S3 object size in single PUT	single upload, PUT object, multipart threshold	5 GB single PUT; 5 TB with multipart upload	Objects over 100 MB should use multipart. Over 5 GB <i>must</i> use multipart upload.
Maximum number of S3 buckets per account	bucket limit, how many buckets, account limit S3	100 buckets by default (can request up to 1,000)	Default is 100 buckets per account — can be increased via Service Quotas.
Maximum VPCs per region	VPC limit, how many VPCs, default quota	5 VPCs per region by default	Default is 5 VPCs per region — can increase via Service Quotas request.
Maximum items in a DynamoDB item	DynamoDB item size, row size limit, 400 KB	400 KB per item	DynamoDB items are capped at 400 KB. Store large objects in S3 and reference from DynamoDB.

Maximum RDS storage	RDS storage limit, max database size, storage autoscaling max	64 TB for most RDS engines; Aurora up to 128 TB auto-scaling	Aurora storage grows automatically in 10 GB increments up to 128 TB.
Maximum Kinesis shard throughput	shard throughput, read write limit, 1 MB per shard	1 MB/s write or 1,000 records/s per shard; 2 MB/s read per shard	Add shards to scale Kinesis. Shard splitting increases capacity; merging reduces cost.
EC2 key pair maximum	key pair limit, how many key pairs	5,000 key pairs per region	Rarely a concern, but key pairs are regional and have a high default limit.
Maximum CloudFormation stacks	stack limit, how many stacks, CF quota	2,000 stacks per account per region by default	Large organizations may hit this — use nested stacks or StackSets across accounts.
Maximum SNS subscriptions per topic	SNS limit, subscribers, fan-out scale	12.5 million subscriptions per topic	SNS scales to massive fan-out. The subscription limit is extremely high by design.
Maximum EBS volume size	EBS size limit, largest volume, storage capacity	64 TiB for gp3/io2; 16 TiB for gp2	gp3 and io2 support up to 64 TiB. gp2 caps at 16 TiB.
Maximum Secrets Manager secret size	secret size, how big a secret, secret limit	65,536 bytes (64 KB) per secret	Secrets Manager is for credentials not large blobs — use S3 for large sensitive files.



Pricing Models & Cost Concepts

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Pay nothing upfront for long-term discount	no upfront, all upfront, partial upfront, RI payment, savings plan payment	Reserved Instance / Savings Plan Payment Options	All Upfront = biggest discount. No Upfront = smallest discount but no capital expense.
Reduce cost for predictable steady-state workload	steady state, always on, predictable, baseline, reserved	Reserved Instances or Savings Plans	For always-on workloads, RIs and Savings Plans save up to 72% vs on-demand.
Reduce cost for flexible but consistent compute	Compute Savings Plan, flexible instance, any instance family, any region	Compute Savings Plans	Compute Savings Plans apply to any EC2 instance, Lambda, and Fargate — most flexible.
Reduce cost for specific instance type	EC2 Instance Savings Plan, specific family, specific region, instance type commitment	EC2 Instance Savings Plans	Locked to a specific instance family and region — higher discount than Compute Savings Plans.
Get the biggest EC2 discount	spot instance, 90% discount, fault tolerant, stateless, batch, interruptible	EC2 Spot Instances	Spot gives up to 90% discount but instances can be reclaimed with 2-minute warning.

Charge based on actual usage	pay per request, per invocation, per GB, consumption model, serverless pricing	Serverless Pricing Model	Lambda, DynamoDB on-demand, S3, SQS, SNS all charge based on actual consumption — no idle cost.
Understand data transfer costs	egress, data transfer out, inter-region, cross-AZ, data transfer pricing	AWS Data Transfer Costs	Inbound is free. Outbound to internet costs. Cross-AZ in same region costs. Cross-region costs.
Avoid data transfer costs between services	same region, VPC endpoint, S3 Gateway endpoint, free data transfer	Same-Region + VPC Endpoints	Traffic between services in the same region using VPC endpoints avoids data transfer charges.
Understand EC2 billing granularity	per second billing, per hour, minimum charge, billing increment	EC2 Per-Second Billing	EC2 Linux instances bill per second with a 1-minute minimum. Windows bills per hour.
Right-size over-provisioned resources	right-sizing, Compute Optimizer, recommendations, instance right-size, over-provisioned	AWS Compute Optimizer	Compute Optimizer analyzes CloudWatch metrics and recommends right-sized instance types.



Networking — Expert Level

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Route traffic between two VPCs in same account	VPC peering, private route, inter-VPC, peering connection	VPC Peering	VPC peering connects two VPCs privately — but it's not transitive. VPC A↔B and B↔C doesn't mean A↔C.
Route traffic through centralized inspection	inspection VPC, centralized firewall, traffic inspection, hub VPC, Gateway Load Balancer	Gateway Load Balancer (GWLB)	GWLB inserts third-party virtual appliances transparently into the traffic path at scale.
Deploy third-party network appliances	virtual appliance, NVA, firewall appliance, IDS/IPS, GWLB, bump-in-the-wire	AWS Gateway Load Balancer	GWLB is designed to scale and manage third-party virtual network appliances (Palo Alto, Fortinet etc.).
Create a private hosted zone for internal DNS	private DNS, internal hostname, VPC DNS, Route 53 private zone, resolve internally	Route 53 Private Hosted Zone	Private hosted zones resolve custom domain names to private IPs within one or more VPCs.
Restrict outbound internet from private EC2	NAT gateway, outbound only, private subnet internet, egress, no inbound	NAT Gateway (in public subnet)	NAT Gateway must be in a public subnet. Route private subnet traffic to NAT Gateway for outbound internet.

Share subnets across accounts	shared VPC, VPC sharing, RAM, Resource Access Manager, centralized VPC	AWS Resource Access Manager (RAM) + VPC Sharing	RAM lets you share subnets from one account's VPC with other accounts in the same Organization.
Multicast traffic in cloud	multicast, one-to-many, Transit Gateway multicast, multicast group	Transit Gateway Multicast	Transit Gateway supports IP multicast — rare feature but testable for video streaming scenarios.
Create redundant Direct Connect links	LAG, Link Aggregation Group, redundant ports, bonded connection, active/active DX	Direct Connect LAG	LAG bundles multiple Direct Connect ports into one logical connection for higher bandwidth and redundancy.
Measure actual network performance	iPerf, network performance, bandwidth test, throughput, enhanced networking	Enhanced Networking (SR-IOV / ENA)	Enable Enhanced Networking for higher throughput and lower latency — required for cluster placement groups.



Security — Expert Level

If the question says...	Look for these keywords	Answer is...	Why this is the clue
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Centrally manage firewall rules across VPCs	Firewall Manager, centralized WAF, security policy, organization firewall, centralized SG	AWS Firewall Manager	Firewall Manager centrally manages WAF rules, Shield protections, and Security Groups across accounts.
Automatically remediate security findings	Security Hub, auto-remediation, finding response, EventBridge + Lambda remediation	Security Hub + EventBridge + Lambda	Security Hub findings trigger EventBridge rules that invoke Lambda to auto-remediate issues.
Enforce encryption in transit	TLS in transit, HTTPS only, enforce TLS, deny HTTP, policy condition	S3 Bucket Policy (aws:SecureTransport) / ACM	Add a bucket policy denying requests where aws:SecureTransport is false to force HTTPS.
Protect against SSRF attacks on EC2 metadata	IMDSv2, metadata v2, session-oriented, SSRF protection, token-based metadata	EC2 IMDSv2	IMDSv2 requires a session token to access instance metadata — protects against SSRF exploits.
Verify software supply chain integrity	code signing, artifact signing, Signer, trusted code, Lambda code signing	AWS Signer	AWS Signer signs and validates code to ensure only trusted code runs in Lambda or IoT.

Detect sensitive data in real time in streams	Macie, stream sensitive data, Kinesis sensitive data, real-time PII, data protection	Amazon Macie / Kinesis Data Streams data protection	Macie monitors S3; Kinesis Data Streams supports data protection policies to detect PII in streams.
Isolate production workloads from dev	separate accounts, account isolation, Organizations, environment isolation, blast radius	AWS Organizations + Separate Accounts	Best practice: separate AWS accounts per environment (dev/staging/prod) for maximum isolation.
Restrict what an IAM role can do in another account	cross-account role, trust policy, external ID, confused deputy, role assumption condition	IAM Role External ID	External ID prevents the confused deputy problem when third parties assume roles in your account.
Prevent accidental public exposure of resources	block public access, S3 block public, account-level block, org SCP, preventive control	S3 Block Public Access + SCP	Enable Block Public Access at account level; use SCP to enforce it across all org accounts.
Manage fleet of IoT devices securely	IoT Core, X.509 certificate, device certificate, IoT policy, thing registry, device auth	AWS IoT Core (Device Authentication)	IoT devices authenticate with X.509 certificates — IoT Core validates and routes MQTT messages.



Compute — Expert Level

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Reduce cold start for Lambda	provisioned concurrency, warm Lambda, pre-initialized, cold start latency, response time	Lambda Provisioned Concurrency	Provisioned concurrency keeps Lambda instances initialized and ready — eliminates cold starts.
Process SQS messages with Lambda	SQS trigger, Lambda event source mapping, batch size, poller, trigger from queue	Lambda SQS Event Source Mapping	Lambda natively polls SQS and invokes in batches — fully managed, no custom polling code needed.
Handle Lambda failures gracefully	DLQ, Lambda destination, async invocation failure, error handling Lambda, on failure	Lambda Destinations / DLQ	Use Lambda Destinations to route successful/failed async invocations to SQS, SNS, or another Lambda.
Run a container image in Lambda	container image Lambda, Docker Lambda, OCI image, 10 GB Lambda, container runtime	Lambda Container Image Support	Lambda supports container images up to 10 GB — larger than the 250 MB zip limit.
Use graviton processors for cost savings	Graviton, ARM, Graviton2 Graviton3, cost savings ARM, 40% cheaper compute	EC2 Graviton / Lambda Graviton	Graviton instances offer up to 40% better price-performance. Lambda

also supports arm64 architecture.

Spot instance interruption handling	interruption notice, 2-minute warning, spot rebalance, Spot interruption, graceful shutdown	EC2 Spot Instance Interruption Handler	Instances get a 2-minute warning before termination — use it to checkpoint and drain connections.
Run HPC tightly coupled workloads	HPC, MPI, tightly coupled, high performance compute, cluster networking, EFA	EC2 with EFA (Elastic Fabric Adapter)	EFA provides OS-bypass network for HPC/ML — lower latency than enhanced networking for MPI workloads.
Schedule capacity in advance	capacity reservation, on-demand capacity reservation, guaranteed capacity, zonal reservation	EC2 On-Demand Capacity Reservations	Reserve capacity in a specific AZ without a term commitment — guarantees capacity is available.
Run Windows workloads with dedicated licensing	dedicated host, BYOL, Windows BYOL, SQL Server BYOL, per-socket, per-core	EC2 Dedicated Hosts	Dedicated Hosts are required for BYOL licensing models based on per-socket or per-core metrics.



Database — Expert Level

If the question says...

Look for these keywords

Answer is...

Why this is the clue

Accelerate DynamoDB reads by 10x	DAX, DynamoDB Accelerator, microsecond, in-memory DynamoDB, read acceleration	Amazon DAX	DAX sits in front of DynamoDB and caches responses — reduces read latency from milliseconds to microseconds.
Shard a relational database horizontally	sharding, horizontal partitioning, distributed SQL, RDS sharding, scale out writes	Application-Level Sharding / Amazon Aurora Sharding	RDS doesn't natively shard — you implement sharding at the application layer or use DynamoDB instead.
Handle burst traffic to DynamoDB	adaptive capacity, hot partition, burst capacity, partition throughput, DynamoDB burst	DynamoDB Adaptive Capacity	DynamoDB automatically redistributes capacity to handle uneven traffic across partitions.
Store session state for web app	session store, Redis ElastiCache, session data, stateless application, sticky sessions alternative	ElastiCache (Redis)	Store sessions in Redis instead of sticky sessions — allows any instance to serve any user.
Replicate database with sub-second lag	Aurora replica, replica lag, read replica latency, Aurora replication	Aurora Replicas	Aurora Replicas share the same storage volume — replica lag is typically under 100 milliseconds.
Convert Oracle database to PostgreSQL	SCT, Oracle migration, heterogeneous, schema conversion, PL/SQL to PL/pgSQL	AWS Schema Conversion Tool (SCT)	SCT converts schema, stored procedures, and functions between different database engines.

Global active-active writes for DynamoDB	global tables, multi-region write, active-active DynamoDB, global table replication	DynamoDB Global Tables	Global Tables replicate writes to all configured regions — all regions can accept writes.
Pause database when not in use	Aurora Serverless auto-pause, scale to zero, dev/test database, variable load	Aurora Serverless v1 Auto-Pause	Aurora Serverless v1 can pause after inactivity and resume on connection — ideal for dev/test.
Query multiple databases with one endpoint	Aurora multi-master, RDS Proxy, distributed query, federated query, Redshift Spectrum	Redshift Spectrum / RDS Proxy	Redshift Spectrum queries S3 + Redshift together. RDS Proxy pools connections to one RDS endpoint.



End-User & Front-End Services

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Build a mobile backend	mobile app, Amplify, GraphQL, AppSync, mobile backend, offline sync, real-time mobile	AWS Amplify + AppSync	Amplify provides full mobile/web backend; AppSync adds real-time GraphQL with offline sync.

Build a GraphQL API	GraphQL, AppSync, query mutation subscription, real-time data, resolvers	AWS AppSync	AppSync is the managed GraphQL service — supports real-time subscriptions and offline sync.
Host a full-stack web application	Amplify Hosting, CI/CD web, full-stack hosting, frontend deploy, Amplify console	AWS Amplify Hosting	Amplify Hosting provides Git-based CI/CD and hosting for web frontends (React, Next.js, Vue).
Send push notifications to mobile devices	SNS mobile push, APNs, FCM, push notification, Pinpoint push, mobile notification	Amazon SNS / Amazon Pinpoint	SNS sends push to APNs/FCM; Pinpoint adds analytics, segmentation, and A/B testing for campaigns.
Send targeted marketing campaigns	Pinpoint, email campaign, SMS campaign, user segmentation, marketing analytics, customer journey	Amazon Pinpoint	Pinpoint is the multi-channel marketing communications service — email, SMS, push, and voice.
Add location features to an app	Amazon Location Service, maps, geofencing, tracking, geocoding, routing, place search	Amazon Location Service	Location Service provides maps, geocoding, tracking, and geofencing without exposing data to third parties.
Build a voice interface or Alexa skill	Alexa, voice skill, VUI, voice interface, Lex voice, custom skill	Amazon Lex / Alexa Skills Kit	Lex builds voice bots on any platform; Alexa Skills Kit builds specifically for Alexa devices.

Stream video to large audiences	IVS, Interactive Video Service, live streaming, video broadcast, Twitch-like, low-latency live	Amazon Interactive Video Service (IVS)	IVS provides managed live streaming with ultra-low latency — Twitch technology for your app.
Add video calling or chat to an app	Chime SDK, video calling, real-time communication, WebRTC, messaging SDK, RTC	Amazon Chime SDK	Chime SDK embeds real-time audio/video/messaging into applications using WebRTC.



IoT & Edge Computing

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Connect IoT devices to AWS	IoT Core, MQTT, device gateway, thing, shadow, IoT policy, device connection	AWS IoT Core	IoT Core is the managed MQTT broker — devices publish/subscribe via MQTT, WebSocket, or HTTPS.
Run logic locally on IoT devices	Greengrass, edge compute, local Lambda, IoT edge, offline processing, local inference	AWS IoT Greengrass	Greengrass extends AWS Lambda and ML inference to edge devices — works offline.

Manage fleet of IoT devices	IoT Device Management, OTA update, fleet indexing, thing groups, bulk registration	AWS IoT Device Management	Device Management handles registration, organization, OTA updates, and remote actions for device fleets.
Store and query IoT time-series data	IoT Analytics, IoT data store, device data history, channeling IoT data	AWS IoT Analytics	IoT Analytics filters, transforms, and stores IoT data in a time-series optimized data store.
Monitor industrial equipment	IoT SiteWise, industrial data, OPC-UA, equipment monitoring, plant floor, asset model	AWS IoT SiteWise	SiteWise collects and organizes data from industrial equipment at scale.
Process data before sending to cloud	edge processing, local compute, Outposts edge, Snowball Edge compute, local inference	Snowball Edge / Greengrass	Snowball Edge has compute for local processing in disconnected environments; Greengrass for connected.



Media & Content Services

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Convert video to	MediaConvert, video transcoding, format	AWS Elemental MediaConvert	MediaConvert is file-based video transcoding —

multiple formats	conversion, MP4 HLS DASH, file-based video		converts source files to streaming-ready formats.
Stream live video	MediaLive, live encoding, live stream, real-time video, contribution encoding, live broadcast	AWS Elemental MediaLive	MediaLive encodes live video streams — ingests contribution feed and outputs to MediaPackage or CDN.
Package and originate video streams	MediaPackage, HLS, DASH, CMAF, just-in-time packaging, origin, DRM	AWS Elemental MediaPackage	MediaPackage packages live video into adaptive bitrate formats and provides the origin for CloudFront.
Store and manage media assets	MediaStore, low-latency object store, media origin, video storage, live media store	AWS Elemental MediaStore	MediaStore is a purpose-built origin store for live video with low latency and high concurrency.
Deliver streaming video globally	CloudFront + MediaPackage, video CDN, HLS delivery, adaptive bitrate, signed cookies	CloudFront (Video Streaming)	CloudFront delivers HLS/DASH streams globally — use signed cookies or signed URLs for access control.



Business Applications

If the question says...	Look for these keywords	Answer is...	Why this is the clue
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Managed contact center	Connect, call center, cloud contact center, IVR, agent, CCP, phone support platform	Amazon Connect	Connect is AWS's cloud contact center — includes IVR, routing, recording, and analytics.
Business email and calendar	WorkMail, managed email, corporate email, IMAP, calendar, AWS email service	Amazon WorkMail	WorkMail is a managed business email and calendar service compatible with Outlook.
Managed desktop as a service	WorkSpaces, virtual desktop, VDI, cloud desktop, DaaS, Windows desktop cloud	Amazon WorkSpaces	WorkSpaces provides cloud-hosted Windows or Linux virtual desktops — no on-prem VDI needed.
Provide access to streaming apps	AppStream 2.0, application streaming, stream desktop app, SaaS streaming, browser-based app	Amazon AppStream 2.0	AppStream streams desktop applications to a browser — no client install required.
Collaborative document editing	WorkDocs, managed file sharing, document collaboration, secure document, Dropbox alternative	Amazon WorkDocs	WorkDocs is a fully managed secure content collaboration and document sharing service.



Well-Architected — Six Pillars Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Improve operational processes	runbooks, playbooks, operational excellence, automate operations, event response, annotate docs	Operational Excellence Pillar	Focus: automate changes, respond to events, define standards for daily ops.
Make workload more secure	least privilege, data protection, incident response, detective controls, security pillar	Security Pillar	Five areas: IAM, detection, infra protection, data protection, incident response.
Increase workload availability	fault isolation, multi-AZ, auto recovery, reliability pillar, quota management, change management	Reliability Pillar	Focus: recover from failures, scale dynamically, manage quotas.
Improve response time and throughput	caching, right instance type, performance efficiency, benchmarking, serverless	Performance Efficiency Pillar	Use the right resource types, sizes, and architectures — experiment frequently.
Reduce unnecessary spend	eliminate waste, cost optimization pillar, right-sizing, use managed services, consumption model	Cost Optimization Pillar	Measure, optimize over time, use elasticity and spot — only pay for what you need.

Reduce environmental impact	carbon footprint, sustainability pillar, energy efficiency, shared infrastructure, minimize waste	Sustainability Pillar	Sixth pillar added in 2021: maximize utilization, use managed services, reduce idle resources.
Identify risks in your architecture	Well-Architected Review, WA Tool, risk, high risk issue, HRI, improvement plan, workload review	AWS Well-Architected Tool	The WA Tool guides structured reviews against all six pillars and identifies HRIs.
Apply best practices automatically	AWS Trusted Advisor, checks, best practice automation, account-level review	AWS Trusted Advisor	Trusted Advisor checks cost, security, fault tolerance, performance, and limits automatically.



Data Movement & Transfer Patterns

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Speed up S3 uploads from far away	Transfer Acceleration, upload speed, CloudFront edge upload, multipart over edge	S3 Transfer Acceleration	Routes uploads through nearest CloudFront edge to AWS backbone — speeds up long-distance uploads.
Sync files incrementally	DataSync, incremental sync, NFS to S3, SMB to	AWS DataSync	DataSync transfers only changed data and can

between on-prem and S3	S3, scheduled transfer, bandwidth throttle		throttle bandwidth to not impact production.
Move exabyte-scale data offline	Snowmobile, 100 PB, truck, 45-foot container, datacenter migration, massive transfer	AWS Snowmobile	Snowmobile is a literal truck with 100 PB capacity — use when Snowball would take too many trips.
Process data locally during transfer	Snowball Edge Compute, local Lambda on Snowball, edge compute, disconnected processing	AWS Snowball Edge (Compute Optimized)	Snowball Edge Compute has EC2 instances and Lambda locally — process data before sending to AWS.
Mirror on-prem volume to AWS for backup	Storage Gateway Volume Gateway, cached volume, stored volume, iSCSI, EBS snapshot	Storage Gateway — Volume Gateway	Cached mode: primary on S3, cached locally. Stored mode: primary on-prem, async backed to S3.
Continuously replicate data between regions	S3 CRR, DynamoDB global tables, Aurora global, RDS cross-region replica, continuous replication	Cross-Region Replication (per service)	Each service has its own CRR mechanism — S3 CRR, DynamoDB Global Tables, Aurora Global Database.
Aggregate data from many sources into one lake	Lake Formation, Glue crawler, S3 data lake, centralize data, data catalog	AWS Lake Formation + Glue	Lake Formation + Glue Crawlers build, secure and catalog a data lake from multiple sources.

Stream database changes to other systems

DMS CDC, database CDC, change data capture, DynamoDB Streams, Aurora Streams, Kafka

DMS CDC / DynamoDB Streams

DynamoDB Streams captures item-level changes. DMS CDC streams relational DB changes to targets.



Encryption Patterns

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Encrypt S3 with AWS-managed keys	SSE-S3, AES-256, default encryption, S3-managed keys, server-side S3	SSE-S3	AWS manages the keys entirely. Simplest option — no extra cost, no KMS API calls.
Encrypt S3 with KMS keys for auditability	SSE-KMS, KMS key, audit key usage, CMK, customer managed key, CloudTrail KMS	SSE-KMS	KMS keys log every use in CloudTrail — use when you need audit trail of who decrypted what.
Encrypt S3 with your own keys	SSE-C, customer-provided key, BYOK S3, client manages key, server encrypts	SSE-C	You provide the key per request — AWS encrypts but never stores your key.

Encrypt data before sending to S3	client-side encryption, encrypt client, SDK encrypt, encrypt before upload	S3 Client-Side Encryption	Encrypt data in your app before uploading — AWS never sees plaintext data.
Encrypt EBS volumes	EBS encryption, KMS EBS, encrypted volume, encrypt at rest EC2	EBS Encryption with KMS	EBS encryption uses KMS — snapshots of encrypted volumes are also encrypted automatically.
Encrypt RDS at rest	RDS encryption, KMS RDS, encrypt database, cannot encrypt existing	RDS Encryption (KMS)	Enable at creation only — to encrypt an existing DB, snapshot → copy encrypted → restore.
Manage your own hardware keys	CloudHSM, dedicated HSM, FIPS 140-2 Level 3, single-tenant HSM, bring your own HSM	AWS CloudHSM	CloudHSM gives dedicated hardware — stricter than KMS, required for FIPS 140-2 Level 3.
Rotate KMS keys automatically	key rotation, annual rotation, automatic rotation, key management lifecycle	KMS Automatic Key Rotation	Enable automatic rotation on customer-managed keys — rotates annually, old versions retained.
Use envelope encryption	envelope encryption, data key, wrapped key, encrypt the key, DEK CEK	AWS KMS Envelope Encryption	KMS generates a data key (DEK) to encrypt data and then encrypts the DEK with a master key (CMK).
Share encrypted snapshots cross-account	encrypted snapshot sharing, KMS key	KMS Key Policy +	Share KMS key policy to the target account AND share the snapshot — both required for

share, cross-account
decrypt, snapshot + key

**Snapshot
Sharing**

cross-account encrypted
snapshots.



Testing & Resilience

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Simulate failures in production	chaos engineering, fault injection, FIS, failure simulation, game day, inject faults	AWS Fault Injection Simulator (FIS)	FIS runs controlled chaos experiments — inject CPU stress, kill instances, throttle APIs.
Test how app behaves under load	load testing, performance test, stress test, distributed load, Artillery, JMeter	AWS Distributed Load Testing (or third-party on EC2)	AWS provides a Distributed Load Testing solution; load test tools like JMeter can run on EC2.
Validate DR plan without traffic impact	DR drill, recovery testing, Route 53 failover test, health check test, pilot light test	Route 53 Health Checks + Failover Test	Test DR by temporarily failing health checks and verifying failover routing works as expected.
Run automated compliance checks continuously	Config rules, continuous compliance, remediation, detect	AWS Config Rules + Conformance Packs	Conformance packs bundle multiple Config rules into a single deployable compliance template.

drift, conformance
pack

Inject latency into
service calls

network latency
injection, FIS latency,
network disruption,
chaos latency

**AWS FIS (Network
Latency Action)**

FIS can inject network
latency on EC2 network
interfaces to test timeout
and retry logic.



Analytics — Expert Level

**If the question
says...**

**Look for these
keywords**

Answer is...

Why this is the clue

Query S3 data
without loading
into database

Athena, serverless SQL,
S3 query, pay per scan,
Parquet ORC, ad-hoc
query

Amazon Athena

Athena scans S3 directly
— use Parquet/ORC
columnar formats to
minimize data scanned
and cost.

Reduce Athena
query costs

partition, columnar
format, Parquet, ORC,
compress, partition
pruning

**Athena
Partitioning +
Columnar
Formats**

Partition data by
date/region and use
Parquet — dramatically
reduces bytes scanned
and cost.

Join streaming
and static data

Kinesis Data Analytics,
stream enrichment,
reference data, join
stream, lookup table

**Kinesis Data
Analytics
(Reference Data)**

Kinesis Data Analytics can
join a streaming data
source with a static
reference table in S3.

Orchestrate complex data pipelines	Step Functions + Glue, MWAA, Airflow, data pipeline orchestration, workflow engine	AWS MWAA (Managed Airflow) / Step Functions	MWAA is managed Apache Airflow — use for complex multi-step data pipeline orchestration.
Analyze CloudTrail logs with SQL	Athena CloudTrail, query API logs, who called what, S3 CloudTrail logs, log analysis SQL	Athena + CloudTrail S3 Logs	Store CloudTrail logs in S3, create an Athena table, query with SQL — powerful for security forensics.
Share datasets externally	Data Exchange, data product, data subscriber, third-party data, publish dataset	AWS Data Exchange	Data Exchange lets providers publish datasets and consumers subscribe — data stays in S3.
Run Apache Spark without managing clusters	Glue Spark, serverless Spark, Glue ETL Spark, Spark job Glue, managed Spark	AWS Glue (Spark Runtime)	Glue runs Spark jobs serverlessly — no cluster to manage, auto-scales, pay per DPU-hour.
Visualize cost and usage in custom BI tool	Cost and Usage Report, CUR, detailed billing, Athena CUR, Redshift CUR	AWS Cost and Usage Report (CUR)	CUR is the most granular billing data — load into Athena or Redshift for custom cost analytics.



Integration Patterns — Expert Level

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Synchronize data between apps without custom code	AppFlow, Salesforce to S3, ServiceNow sync, no-code integration, data flow	Amazon AppFlow	AppFlow provides no-code bidirectional data flows between SaaS and AWS — runs on schedule or event.
Handle back-pressure in streaming pipeline	SQS buffer, queue depth, auto scaling trigger, backpressure, consumer lag	SQS + Auto Scaling (queue depth alarm)	Use CloudWatch SQS ApproximateNumberOfMessages to trigger Auto Scaling — classic back-pressure pattern.
Pattern match and route events	EventBridge rule, event pattern, filter events, route by attribute, content-based routing	EventBridge Event Patterns	EventBridge rules match event content and route to targets — filter by source, detail-type, or any field.
Call external HTTP endpoint from workflow	Step Functions HTTP task, API Gateway integration, EventBridge API destination, webhook	Step Functions HTTP Task / EventBridge API Destination	Step Functions can call any HTTPS endpoint directly; API Destination sends EventBridge events to external APIs.
Ensure at-least-once message delivery	SQS standard, at-least-once, idempotency, deduplication,	SQS Standard Queue	Standard queues guarantee at-least-once delivery — design consumers to be idempotent.

consumer
idempotent

Process messages in strict order	FIFO queue, SQS FIFO, ordered processing, sequence, message group ID	SQS FIFO Queue	FIFO queues guarantee strict ordering within a message group — use MessageGroupId to partition.
Trigger workflow on database change	DynamoDB Streams + Lambda, database trigger, item change event, stream processor	DynamoDB Streams + Lambda	Streams capture every item change — Lambda processes the stream in near real-time as an event source mapping.
Build event-driven microservices	EventBridge, event bus, loosely coupled, pub-sub microservices, domain events	Amazon EventBridge	EventBridge is the backbone for event-driven microservices — services emit events, others subscribe.



Reliability & High Availability Patterns

If the question
says...

Look for these
keywords

Answer is...

Why this is the clue

Automatically replace unhealthy instances	Auto Scaling health check, EC2 health check, ELB health check, replace instance, self-healing	Auto Scaling Group Health Checks	ASG replaces instances that fail EC2 or ELB health checks — key self-healing mechanism.
Maintain minimum capacity during AZ failure	desired capacity, min capacity, cross-AZ, AZ rebalance, Auto Scaling AZ	Auto Scaling Multi-AZ with min/desired capacity	Spread ASG across 3 AZs with min=3 — if one AZ fails, ASG launches replacements in remaining AZs.
Detect and route around unhealthy endpoints	Route 53 health check, failover record, active-passive, health endpoint, failover routing	Route 53 Health Checks + Failover Routing	Route 53 health checks poll an endpoint and switch traffic to the failover record if unhealthy.
Prevent a single bad deployment from causing outage	rolling deployment, canary, blue/green, deployment strategy, CodeDeploy rollback	Blue/Green or Canary Deployment	Blue/green: switch all traffic at once with easy rollback. Canary: shift small % first, then full.
Recover database in case of full region failure	Aurora Global Database failover, promote secondary, managed failover, RTO < 1 min	Aurora Global Database Managed Failover	Aurora Global can promote a secondary region to primary in under 1 minute for region-level DR.
Protect against deletion of critical data	S3 MFA delete, S3 Object Lock, RDS deletion protection, termination protection	Deletion Protection (per service)	Enable deletion protection on RDS, EC2 termination protection, and S3 Object Lock to prevent accidents.

Test that backups actually work	backup restore test, recovery test, DR test, validate backup, test restore	AWS Backup Restore Testing (or manual restore)	Regularly test restores — a backup you've never tested is not a real backup. FIS can also test this.
Implement circuit breaker for microservices	circuit breaker, App Mesh, service mesh, retry, timeout, bulkhead, resilience	AWS App Mesh	App Mesh implements service mesh — provides retries, timeouts, circuit breakers between microservices.
Rate-limit API calls from a client	throttle, rate limit, API Gateway usage plan, burst, quota per API key, 429	API Gateway Usage Plans + API Keys	Usage Plans enforce per-client rate limits and quotas — returns 429 when exceeded.



Containers — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Run a container task once on a schedule	ECS scheduled task, cron container, EventBridge ECS, one-off task, scheduled container	ECS Scheduled Tasks (via EventBridge)	EventBridge triggers ECS RunTask on a schedule — serverless cron replacement using containers.

Pass secrets to containers securely	ECS secrets, container env secrets, Secrets Manager integration, SSM secrets ECS	ECS + Secrets Manager / Parameter Store	ECS can inject Secrets Manager or SSM Parameter Store values as environment variables at task launch.
Auto-scale containers based on load	ECS Service Auto Scaling, target tracking, step scaling, ALB request count, container scaling	ECS Service Auto Scaling	Scale ECS tasks based on CPU, memory, or ALB request count per target — just like EC2 ASG.
Use spot capacity for containers	Fargate Spot, ECS Spot, cost-effective containers, interruptible containers, spot task	Fargate Spot	Fargate Spot runs containers on spare capacity at up to 70% discount — can be interrupted.
Deploy containers across multiple regions	multi-region ECS, global container, ECR replication, cross-region container, global ECS	ECR Cross-Region Replication + multi-region ECS	Replicate ECR images to each region and deploy ECS services independently per region.
Monitor container resource usage	Container Insights, ECS metrics, task CPU memory, container performance, X-Ray containers	CloudWatch Container Insights	Container Insights provides CPU, memory, network, and disk metrics for ECS and EKS tasks.
Run Windows containers on AWS	Windows container, Windows ECS, Windows Fargate, .NET container, IIS container	ECS with Windows AMI / Fargate Windows	ECS supports Windows containers on Windows EC2 AMIs; Fargate supports Windows containers too.

Manage
Kubernetes
add-ons

EKS add-on, CoreDNS,
kube-proxy, VPC CNI,
EKS managed add-on,
cluster component

EKS Managed Add-ons

EKS Managed Add-ons
keep CoreDNS, VPC CNI,
and kube-proxy updated
and compatible
automatically.



Serverless — Deep Dive

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Build serverless app framework	SAM, Serverless Application Model, sam deploy, sam build, serverless template, transform	AWS SAM (Serverless Application Model)	SAM is a CloudFormation extension with shorthand for Lambda, API Gateway, DynamoDB, and events.
Share serverless apps with the community	Serverless Application Repository, SAR, publish app, discover serverless, deploy Lambda app	AWS Serverless Application Repository	SAR lets you publish and share SAM applications — also find pre-built serverless components.
Run SQL on serverless data	Athena, serverless query, no cluster, S3 SQL, pay per query, schema on read	Amazon Athena	Athena is fully serverless SQL — no cluster, no maintenance, pay only for data scanned.

Serverless graph database	Neptune Serverless, graph serverless, auto-scale graph, variable graph workload	Amazon Neptune Serverless	Neptune Serverless auto-scales graph database capacity — pay per NCU-hour used.
Serverless NoSQL document store	DocumentDB Serverless, variable document DB, auto-scale document	Amazon DocumentDB Serverless	DocumentDB Serverless scales compute capacity automatically for variable MongoDB-compatible workloads.
Serverless cache	ElastiCache Serverless, auto-scale cache, serverless Redis, serverless Memcached	ElastiCache Serverless	ElastiCache Serverless provisions and scales cache capacity automatically — no cluster management.
Run code in response to HTTP without API Gateway	Lambda Function URL, direct HTTPS Lambda, Lambda URL, no API Gateway, invoke Lambda HTTPS	Lambda Function URLs	Lambda Function URLs provide a dedicated HTTPS endpoint for a Lambda without API Gateway.
Orchestrate long-running serverless workflows	Step Functions, Express Workflow, Standard Workflow, async orchestration, saga, compensating	AWS Step Functions	Standard Workflows support up to 1 year duration; Express Workflows for high-volume short runs.



Search & Discovery

If the question says...	Look for these keywords	Answer is...	Why this is the clue
Full-text search across data	OpenSearch, Elasticsearch, full-text search, search index, search relevance, Kibana	Amazon OpenSearch Service	OpenSearch provides full-text search, log analytics, and Kibana-compatible dashboards.
Product search with ranking	Kendra, intelligent search, enterprise search, document search, FAQ search, ML search	Amazon Kendra	Kendra is ML-powered enterprise search — searches across S3, SharePoint, Confluence, RDS.
Find data across your AWS account	AWS Glue Data Catalog, metadata store, table definition, schema discovery, crawler	AWS Glue Data Catalog	Glue Data Catalog is the central metadata repository — Athena, EMR, and Redshift all query it.
Discover and catalog sensitive data	Macie, data discovery, sensitive data catalog, PII discovery, S3 scan, data inventory	Amazon Macie	Macie inventories S3 buckets and identifies buckets with sensitive data or misconfigured access.
Search for resources across AWS accounts	Resource Explorer, cross-account resource search, find resources, multi-account search	AWS Resource Explorer	Resource Explorer indexes and searches AWS resources across regions and accounts from one place.

Tag-based
resource
discovery

Tag Editor, find by tag,
bulk tagging, resource
group, tagging query

**AWS Tag Editor
/ Resource
Groups**

Tag Editor finds and tags
resources across services;
Resource Groups organizes
by shared tags.



Messaging Patterns — Tricky Distinctions

If the question says...	Look for these keywords	Answer is...	Why this is the clue
One producer, one consumer, async	SQS, point-to-point, queue, one consumer, poll, decoupled	Amazon SQS	SQS is point-to-point — one message is consumed by one consumer. Use for async task offloading.
One producer, many consumers simultaneously	SNS, fan-out, topic, publish-subscribe, multiple endpoints, broadcast	Amazon SNS	SNS pushes one message to all subscribers simultaneously — use for fan-out.
Fan-out AND persist for each consumer	SNS + SQS fan-out, SNS to multiple SQS, durable fan-out, per-consumer queue	SNS → multiple SQS queues	Classic pattern: SNS fans out, each downstream service has its own SQS queue for durability.
Real-time ordered event log	Kinesis Data Streams, event log, ordered, partition key,	Amazon Kinesis Data Streams	Kinesis maintains ordering per shard and allows multiple consumers to read independently.

consumer group,
replay

Trigger Lambda
from DynamoDB
changes

DynamoDB Streams,
stream + Lambda,
event source
mapping, item change
trigger

**DynamoDB
Streams +
Lambda ESM**

Lambda's Event Source
Mapping polls DynamoDB
Streams — no custom
poller needed.

Open protocol
messaging (AMQP,
STOMP, MQTT)

ActiveMQ, RabbitMQ,
MQ, open standard,
AMQP, JMS, STOMP,
wire protocol

Amazon MQ

MQ is for existing apps
using open messaging
protocols — migrate
without rewriting message
code.

IoT
publish-subscribe at
scale

MQTT, IoT broker,
device pub-sub, IoT
Core, topic filter, QoS

AWS IoT Core

IoT Core is a managed
MQTT broker — optimized
for millions of low-power
devices.